

ARIMA - BASED FORECASTING OF MALAYSIA'S ELECTRICAL AND
ELECTRONICS EXPORTS

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LIST OF ABBREVIATIONS

ADF	-	Augmented Dickey-Fuller (test)
AIC	-	Akaike Information Criterion
ACF	-	Autocorrelation Function
ARDL	-	Autoregressive Distributed Lag
ARIMA	-	Autoregressive Integrated Moving Average
BIC	-	Bayesian Information Criterion
XGBoost	-	Extreme Gradient Boosting
CRISP-DM	-	Cross-Industry Standard Process for Data Mining
CPI	-	Consumer Price Index
CV	-	Coefficient of Variation
DOSM	-	Department of Statistics Malaysia
E&E	-	Electrical and Electronics
EDA	-	Exploratory Data Analysis
FRED	-	Federal Reserve Economic Data
GDP	-	Gross Domestic Product
IPI	-	Industrial Production Index
KPI	-	Key Performance Indicator
MAE	-	Mean Absolute Error
MATRADE	-	Malaysia External Trade Development Corporation
MITI	-	Ministry of Investment, Trade and Industry
ML	-	Machine Learning
MSE	-	Mean Squared Error
MYR	-	Malaysian Ringgit
PACF	-	Partial Autocorrelation Function
RBF	-	Radial Basis Function (kernel)
RF	-	Random Forest
RMSE	-	Root Mean Squared Error
SARIMA	-	Seasonal Autoregressive Integrated Moving Average
SVR	-	Support Vector Regression

LIST OF SYMBOLS

p	-	Order of the autoregressive (AR) term
d	-	Order of differencing
q	-	Order of the moving average (MA) term
P	-	Seasonal autoregressive order (SARIMA)
D	-	Seasonal differencing order (SARIMA)
Q	-	Seasonal moving average order (SARIMA)
s	-	Length of seasonal cycle (SARIMA, e.g. 12 for monthly data)
R^2	-	Reynold Number
α	-	Significance Level

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CHAPTER 1

INTRODUCTION

1.1 Overview

Malaysia's Electrical and Electronics (E&E) sector is a major contributor to the country's export earnings, and short-term visibility into its export performance is valuable to policymakers, trade analysts, and industry stakeholders alike. This chapter introduces the practical forecasting problem this study addresses: the need for accurate, short-horizon (one-, two-, and three-month-ahead) forecasts of Malaysia's monthly E&E export values.

This study develops and validates an ARIMA time-series model as the primary forecasting method, since ARIMA is a well-established statistical approach capable of forecasting a series directly from its own historical trend and autocorrelation structure. Alongside the forecasting model, this study also examines the historical relationship between E&E exports and three macroeconomic indicators — CPI inflation, the MYR/USD exchange rate, and the U.S. Industrial Production Index — to provide broader descriptive context on the economic conditions associated with export performance. As supplementary work, machine-learning approaches (Random Forest, Support Vector Regression, and XGBoost) were also implemented and compared against the ARIMA model; these results are presented in the Appendix rather than forming part of the study's main forecasting methodology.

The remainder of this chapter presents the problem background, problem statement, research questions and objectives, research significance, and research scope that together define and delimit this study.

1.2 Problem Background

The Electrical and Electronics (E&E) industry remains one of Malaysia's most important economic sectors, contributing approximately 6.9% to national GDP in 2021 and around 32% of total manufacturing sales value in 2022 (Sheng and Ismail, 2025). Malaysia's overall exports reached a record RM1.607 trillion in 2025, with technology-driven products, including E&E, playing a substantial role (MATRADE, 2025). Because the sector is deeply embedded in global electronics supply chains, its export performance is highly sensitive to shifts in domestic economic conditions and external demand, and these shifts can occur within short time windows that are difficult to anticipate using conventional planning tools.

Accurate one-, two- and three-month-ahead forecasts of E&E export performance are valuable to several groups of decision-makers. Policymakers and government agencies rely on timely forecasts to anticipate emerging risks and calibrate trade and economic policy responses before conditions deteriorate further. Trade analysts and financial institutions use short-term forecasts to assess market conditions and support risk assessment and lending decisions. E&E exporters and businesses depend on near-term visibility into export performance to plan production, manage inventory and allocate resources efficiently.

Existing forecasting approaches for Malaysia's E&E sector present two related limitations for this practical problem. First, prior Malaysia-specific forecasting work has tended to rely on quarterly rather than monthly data, reducing both the resolution and the number of observations available for modelling (Sheng and Ismail, 2025). Second, econometric studies examining Malaysia's electronics and manufacturing exports have generally used cointegration and ARDL frameworks designed to estimate long-run relationships between exports and macroeconomic conditions such as exchange rate and inflation these studies are valuable for understanding why exports move over the long run, but they are not designed to produce operational, short-horizon monthly forecasts (Wong, 2008; Baharin, Shahperi and Guangqin, 2023; Hakim, 2024).

This study addresses the practical forecasting gap by retaining monthly E&E export data across a 35-year period (January 1990 to March 2025) and developing an ARIMA time-series model, which forecasts future export values directly from the series' own historical pattern — its trend, momentum, and autocorrelation structure — without requiring external predictor data at forecast time. Separately, and to provide broader context for understanding the series, this study also examines the descriptive relationship between E&E exports and three macroeconomic indicators — CPI inflation, the MYR/USD exchange rate, and the U.S. Industrial Production Index — through correlation and trend analysis. This relationship analysis is exploratory and contextual: it helps explain what else moves alongside E&E exports, but it does not feed into the ARIMA forecasting model itself, since ARIMA is univariate and forecasts each future value from the series' own past observations.

1.3 Problem Statement

Malaysia's E&E export values fluctuate in response to short-term shifts in both domestic and external economic conditions, yet forecasting these fluctuations accurately remains difficult. Because the sector is closely tied to global electronics demand and international supply chains, its export performance can shift within short time windows in ways that are not always captured by forecasting approaches built on lower-frequency or narrowly scoped data.

Existing research has approached this problem from several directions, but each carries a specific limitation. Malaysia-specific forecasting work has concentrated on E&E sales values and their contribution to GDP rather than export values directly, and where monthly data existed, it was aggregated into quarterly observations to match quarterly GDP, reducing both the resolution and the number of observations available for modelling (Sheng and Ismail, 2025). Separately, econometric studies have established that exchange rate and related macroeconomic conditions influence Malaysia's electronics and manufacturing exports (Wong, 2008; Baharin, Shahperi and Guangqin, 2023; Hakim, 2024), but these studies rely on cointegration and ARDL frameworks designed to estimate long-run structural relationships rather than to generate operational short-term forecasts. Time-series forecasting methods such as

ARIMA are well established for short-term forecasting of economic series generally, and have demonstrated strong performance in export-forecasting studies in other national contexts (Nooraeni *et al.*, 2022; Qiu, 2022), yet no existing study has applied and rigorously validated an ARIMA model specifically for Malaysia's monthly E&E export series across multiple short-term forecasting horizons.

This gap the absence of a properly identified, diagnostically validated ARIMA forecasting model for Malaysia's monthly E&E exports at the one-, two-, and three-month horizons relevant to policymakers, trade analysts, and E&E industry stakeholders motivates the present study. To address it, this study develops an ARIMA time-series model for Malaysia's monthly E&E export values using data from January 1990 to March 2025, following a structured model-identification process (stationarity testing, differencing, autocorrelation analysis, and order selection) to produce a robust and well-validated forecasting model. The historical relationship between E&E exports and selected macroeconomic indicators CPI inflation, the MYR/USD exchange rate, and the U.S. Industrial Production Index is also examined to provide contextual insight into the series' broader economic drivers, complementing the ARIMA analysis.

1.3.1 Research Questions

This following research questions in this study are:

Research Question 1: What patterns, trends and relationships exist between Malaysia's monthly E&E export values, historical time-series characteristics and the selected macroeconomic indicators?

Research Question 2: How can an ARIMA time-series model be developed, validated, and used to forecast Malaysia's monthly E&E export values at one-, two- and three-month-ahead horizons?

Research Question 3: How can an interactive Power BI dashboard present historical E&E export trends, model performance, and one-, two- and three-month-ahead forecast results to support evidence-based decision-making?

1.4 Project Goal and Objectives

The goal of this project is to develop and evaluate an ARIMA-based time-series forecasting model for Malaysia's Electrical and Electronics (E&E) exports using monthly historical export data. The study focuses on identifying the historical patterns, trends, autocorrelation structure, and seasonal characteristics of E&E exports and using these characteristics to develop a suitable ARIMA model for short-term forecasting. The model will generate one-, two-, and three-month-ahead forecasts of Malaysia's E&E export values. The forecasting performance of the developed model will be evaluated using appropriate forecast accuracy measures and diagnostic analysis to assess its reliability and suitability for short-term forecasting. In addition, the historical relationships between E&E exports and selected macroeconomic indicators, namely CPI inflation, the MYR/USD exchange rate, and the U.S. Industrial Production Index (IPI), will be examined to provide contextual understanding of factors associated with E&E export performance. The resulting historical trends, model performance and forecast results will be presented through an interactive Power BI dashboard to support evidence-based decision-making.

The objectives of this project are:

Objective 1: To analyse the patterns, trends and relationships between Malaysia's monthly E&E export values, historical time-series characteristics and the selected macroeconomic indicators.

Objective 2: To develop, validate and evaluate an ARIMA time-series model for forecasting Malaysia's monthly E&E export values at one-, two- and three-month-ahead horizons.

Objective 3: To develop an interactive Power BI dashboard for presenting historical E&E export trends, model performance and one-, two- and three-month-ahead forecast results to support evidence-based decision-making.

1.5 Research Significance

This study contributes to short-term forecasting of Malaysia's Electrical and Electronics (E&E) exports by developing and validating an ARIMA time-series model using 35 years of monthly export data from January 1990 to March 2025. The use of monthly observations provides a higher-resolution view of E&E export movements and addresses the data-frequency limitation identified in previous Malaysia-specific studies that relied on quarterly data. By modelling the historical structure of the E&E export series, including its trend, autocorrelation and seasonal characteristics, the study provides a statistically grounded approach for generating short-term export forecasts.

The study also examines the historical relationships between E&E exports and three selected macroeconomic indicators, namely CPI inflation, the MYR/USD exchange rate and the U.S. Industrial Production Index (IPI). These indicators are analysed to provide contextual understanding of the broader economic conditions associated with Malaysia's E&E export performance. Although they are not used as input variables in the ARIMA forecasting model, their relationships with E&E exports provide additional insight into the economic environment surrounding the export series.

The one-, two-, and three-month-ahead forecasts generated by the ARIMA model may provide useful information for policymakers, trade analysts, financial institutions and E&E businesses. Policymakers may use the forecasts to monitor potential short-term changes in export performance and support trade-related planning. Trade analysts and financial institutions may use the information to assess near-term economic and market conditions, while E&E businesses may consider the forecasts when planning production, inventory and resource allocation.

The study also contributes a practical means of communicating the forecasting results through an interactive Power BI dashboard. The dashboard presents historical E&E export trends, relevant analytical findings, ARIMA model performance and forecast results in an accessible visual format. This allows users to interact with the information and obtain a clearer understanding of both historical export behaviour and expected short-term movements.

1.6 Research Scope

The following scopes are included in this research:

This study focuses exclusively on forecasting Malaysia's Electrical and Electronics (E&E) export values at the aggregate national level.

The study uses monthly historical E&E export data covering the period from January 1990 to March 2025. The ARIMA model is developed using the historical E&E export series to generate one-, two-, and three-month-ahead forecasts.

The analysis examines three selected macroeconomic indicators, namely CPI inflation, the MYR/USD exchange rate and the U.S. Industrial Production Index (IPI), to investigate their historical relationships with E&E exports and provide contextual economic interpretation. These indicators are not used as input variables in the ARIMA forecasting model. Other potentially relevant economic indicators are outside the scope of this study due to data availability and frequency considerations.

1.7 Project Stakeholders

The ARIMA forecasting model and the accompanying Power BI dashboard developed in this project are intended to serve four stakeholder groups, each of whom has a distinct interest in the short-term direction of Malaysia's Electrical and Electronics (E&E) exports. These groups are identified on the basis of their potential use of the forecasting outputs and the visual components presented in the dashboard.

Agencies such as the Ministry of Investment, Trade and Industry (MITI) and the Malaysia External Trade Development Corporation (MATRADE) are responsible for trade planning and export promotion. The dashboard is useful to them because it consolidates the long-run historical export series (January 1990–March 2025), the one-, two-, and three-month-ahead ARIMA forecasts, and the forecast-versus-actual comparison in a single view. This allows early signals of export slowdown or recovery to be identified ahead of the official trade statistics release, supporting more timely policy and trade-planning decisions.

Firms operating in the E&E sector require short-horizon visibility to plan production volumes, inventory levels, and procurement schedules. The dashboard is useful to them because the one-, two-, and three-month-ahead forecasts are presented together with the 95% forecast intervals, so the expected export value and the plausible range around it can both be read directly. The interval width communicates how much confidence should be placed in each horizon, which supports more realistic short-term operational planning than a single point estimate would allow.

Investors and analysts tracking Malaysia's E&E sector need to interpret sector momentum quickly. The dashboard is useful to them because it displays the historical export trend alongside the forecast trajectory and the selected macroeconomic indicators CPI inflation, the MYR/USD exchange rate, and the U.S. Industrial Production Index which are provided as descriptive economic context. Together with the reported accuracy metrics (MAE, RMSE, and R^2), this allows the sector outlook and the reliability of the forecast supporting it to be assessed at the same time.

Researchers studying Malaysia's export behaviour or time-series forecasting methods benefit from the transparency of the reported results. The dashboard is useful to them because it exposes the selected model specification, ARIMA (2,1,2), together with its walk-forward evaluation metrics across all three horizons and its out-of-sample validation outcome. This provides a documented and reproducible performance benchmark against which alternative specifications, such as SARIMA or machine-learning approaches, can be compared in subsequent work.

A limitation applying to all four groups should be acknowledged: no formal stakeholder consultation or expert evaluation was conducted during this study. The stakeholder needs described above are therefore inferred from the documented decision-making requirements of each group and from the information the dashboard is designed to communicate, rather than established through primary engagement.

1.8 Thesis Organisation

The thesis consists of seven chapters, with each chapter covering a specific aspect of the research.

Chapter 1 – Introduction: Introduces the research background and problem related to forecasting Malaysia’s E&E exports. It presents the problem statement, research questions, project goal and objectives, research significance, research scope, project stakeholders, and thesis organisation.

Chapter 2 – Literature Review: Reviews existing literature related to Malaysia’s E&E industry, E&E exports and export forecasting. The chapter examines relevant macroeconomic factors, time-series forecasting approaches, and previous applications of ARIMA and selected machine-learning methods. The reviewed studies are compared to identify their limitations and establish the research gap addressed by this study.

Chapter 3 – Research Methodology: Describes the methodology adopted for the study based on the CRISP-DM framework. The chapter explains the research design, data collection and preparation, exploratory analysis, ARIMA model development, model identification, parameter estimation, diagnostic checking, forecasting and evaluation procedures. The supplementary machine-learning comparison is also described where applicable.

Chapter 4 – Exploratory Data Analysis: Presents the results of data preparation and exploratory data analysis. The chapter examines the structure, distribution, historical trends, seasonal behaviour, autocorrelation and relationships between

Malaysia's E&E export values and the selected macroeconomic indicators. These findings provide the empirical basis for the subsequent ARIMA model development.

Chapter 5 – Implementation: Describes the implementation of the ARIMA forecasting model, including model identification, fitting, diagnostic checking and generation of one-, two-, and three-month-ahead forecasts. The chapter also describes the implementation of the supplementary Random Forest, SVR and XGBoost models where applicable, as well as the development and implementation of the Power BI dashboard.

Chapter 6 – Results and Discussion: Presents and discusses the results obtained from the implemented forecasting models. The chapter evaluates the forecasting performance for the one-, two-, and three-month-ahead horizons and discusses the findings in relation to the research questions and objectives. The historical patterns, forecasting results, model performance and dashboard outputs are also interpreted and discussed.

Chapter 7 – Conclusion and Recommendations: Summarizes the main findings of the study and explains how the research questions and objectives have been addressed. The chapter discusses the contributions and limitations of the study and provides recommendations for future research.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

This chapter critically reviews existing literature related to forecasting Malaysia's Electrical and Electronics (E&E) export values. The review examines the characteristics of Malaysia's E&E export sector, macroeconomic factors that influence export performance, and time-series forecasting approaches — with particular attention to ARIMA (Autoregressive Integrated Moving Average), the primary forecasting method adopted in this study. The literature is compared in terms of data frequency, forecasting target, predictor variables, geographical context, forecasting horizon, and modelling approach. This comparison is used to identify limitations in existing research and establish the research gap addressed by the present study: the absence of a properly identified, diagnostically validated ARIMA model for Malaysia's monthly E&E export series.

2.2 Problem Domain

The problem domain of this study is the forecasting of Malaysia's Electrical and Electronics (E&E) export values. The E&E industry is an important component of Malaysia's manufacturing and export sector and includes a broad range of products such as semiconductors, electronic components, telecommunications equipment and other electronic products. Malaysia's participation in global E&E value chains has made the sector an important part of the country's international trade activities. The industry's performance is also closely connected to developments in international markets and changes in global demand for electronic and semiconductor-related products (Sheng and Ismail, 2025).

The E&E sector operates within a highly interconnected international environment, where changes in domestic economic conditions and external market activity can influence export performance. Malaysia's E&E exports are exposed to international demand because the industry is integrated into global production and supply chains. Previous research on Malaysia's E&E exports has identified major trading markets, including the United States, Singapore, Japan and Hong Kong, demonstrating the sector's dependence on international markets (Muhammad and Yaacob, 2009). This international exposure makes short-term changes in economic and industrial conditions relevant to the forecasting of Malaysia's E&E export values.

These characteristics make E&E export performance potentially sensitive to changes in both domestic macroeconomic conditions and external industrial demand. These characteristics make both the historical behaviour of E&E exports and relevant economic indicators important for understanding Malaysia's E&E export performance. In this study, the historical export series is used as the basis for ARIMA forecasting, while the selected macroeconomic indicators are examined separately to provide contextual understanding of export movements.

2.2.1 Malaysia's E&E Industry

The Electrical and Electronics (E&E) industry in Malaysia includes semiconductors, electronic components, telecommunication equipment, electrical machine and other advanced electronic products (Sheng and Ismail, 2025). The industry has become an integral part of the manufacturing-based economy of Malaysia in the last 30 years since the country shifted from being a commodity-based economy to a manufacturing-based export economy through its industrialisation policies such as import-substitution industrialisation in the 1960s, export-oriented industrialisation in the 1970s, and various successive Industrial Master Plans in the 1980s and 1990s (Wong, 2008). Foreign direct investment inflows, especially from the United States, Japan, Singapore and the European Union, played a significant part in the growth and development of the sector, as shown in subsequent research which identified the exchange rate, income tax, financial development and economic risk as important

determinants of FDI in the Malaysia E&E sector specifically (Tang, Yip and Ozturk, 2014).

Within the broader manufacturing sector, the E&E industry has become Malaysia's leading non-resource-based export-oriented industry, representing the largest single component of manufactured exports in terms of output, employment, and export value (Muhammad and Yaacob, 2009). Malaysia's participation in the global E&E value chain has positioned the country as a significant international player, with the industry contributing approximately 6.9% to national GDP in 2021 and accounting for around 32% of total manufacturing sales value by 2022 (Sheng and Ismail, 2025). The industry's deep integration into international supply chains means that Malaysia's E&E producers do not operate in isolation from external economic conditions rather, the sector's performance is directly linked to global demand cycles, particularly rising demand for semiconductor-related products driven by electric vehicles, artificial intelligence, and other emerging technologies (Sheng and Ismail, 2025).

2.2.2 Malaysia's E&E Export Performance

Malaysia's E&E exports have historically been concentrated in a small number of major international markets. Early disaggregated analysis identified the United States, Singapore, Japan, and Hong Kong as Malaysia's principal E&E export destinations (Muhammad and Yaacob, 2009), reflecting the sector's exposure to demand conditions in these economies. Comparative competitiveness analysis using Constant Market Share and Revealed Comparative Advantage frameworks found that in the early 1990s, Malaysia's E&E exports were genuinely competitive in these four markets; however, in later periods, continued export growth was found to be driven more by rising global import demand (a structural effect) than by improvements in Malaysia's underlying competitiveness (Muhammad and Yaacob, 2009).

More recent trade data confirms the sector's continued significance: Malaysia's exports reached a record RM1.607 trillion in 2025, with growth underpinned by record shipments to major trading partners including ASEAN, the United States, Taiwan, and the European Union, reflecting continued integration into technology-driven global

supply chains (MATRADE, 2025). At the same time, export performance has proven sensitive to external shocks (Thorbecke, 2024) documents that Malaysia's semiconductor-related firms were among the sectors most negatively affected during the recent period of overlapping global crises, including the COVID-19 pandemic and associated supply chain disruptions, underscoring the sector's vulnerability to conditions outside Malaysia's direct control.

Despite this recognised sensitivity to both domestic and external economic conditions, Malaysia-specific research forecasting E&E export performance directly rather than sales value or GDP contribution remains limited, a gap discussed further in Section 2.6.

2.3 Macroeconomic Factors Influencing E&E Exports

2.3.1 Exchange Rate

The exchange rate has been consistently identified as a significant determinant of Malaysia's electronics and manufacturing export performance. Wong, in a disaggregated study of Malaysia's top five electronics export categories by SITC product group, found a long-run cointegrating relationship between export demand and relative prices, with estimated price elasticities exceeding one for most product categories with the notable exception of semiconductor exports, where price appeared less influential than for other electronics categories. This suggests that exchange-rate-driven price competitiveness affects different E&E sub-sectors unevenly (Wong, 2008).

Similarly it confirms the exchange rate's role using Autoregressive Distributed Lag (ARDL) analysis of Malaysia's manufacturing exports (of which E&E products contributed 30.2% in 2022) over 1990–2021, finding a long-run cointegrating relationship between exchange rate movements and manufacturing export value (Muhammad and Yaacob, 2009). It also reinforces this finding specifically in terms of volatility, concluding that exchange rate volatility negatively affects Malaysia's export volumes and revenues, and that stable exchange rate conditions are important for

maintaining export competitiveness (Hakim, 2024). Taken together, this literature establishes exchange rate not merely as a theoretically relevant variable but as an empirically confirmed determinant of Malaysia's E&E and manufacturing export performance specifically.

2.3.2 Inflation

Inflation is theorised to affect export performance primarily through its effect on production costs: rising domestic prices increase the cost of producing export goods, which can reduce price competitiveness relative to foreign producers (Baharin, Shahperi and Guangqin, 2023). While less extensively tested in the Malaysia-specific E&E literature than exchange rate, inflation has been incorporated as a control variable in broader manufacturing export studies (Baharin, Shahperi and Guangqin, 2023).

Recent machine-learning research on inflation dynamics provides additional justification for how inflation should be modelled in a forecasting context. Pratap, Pawar and Sengupta, applying explainable machine learning to inflation forecasting in an emerging-market context, demonstrate that inflation relationships are frequently non-linear, involving thresholds and interaction effects between variables that traditional linear models struggle to capture. This finding supports the use of non-linear machine-learning approaches such as those applied in the present study over linear econometric specifications when inflation is included as a predictor of export behaviour (Sengupta, Pratap and Pawar, 2025).

2.3.3 US Industrial Production Index

The U.S. Industrial Production Index (IPI) has been used in prior research as a proxy for external industrial demand affecting a trading partner's exports. It was also directly incorporated U.S. IPI, alongside exchange rate and GDP, as one of six variables used to forecast China's export value to the United States, finding that models incorporating this indicator produced satisfactory forecasting performance across multiple modelling approaches. This precedent supports the use of a similar external-

demand proxy for Malaysia's E&E exports, given the sector's substantial trade exposure to U.S. demand (Qiu, 2022).

Further support comes from Kumar and Ahmad, who examine the forecasting relationship between U.S. and Chinese industrial production and industrial production across G20 economies. Their findings indicate that incorporating U.S. industrial production data measurably improves the forecast accuracy of other countries' industrial production, particularly for developed and trade-linked economies, with the effect concentrated in economies most integrated into U.S.-linked supply chains (Kumar and Ahmad, 2024).

2.4 Time-Series Forecasting Approaches: ARIMA and Related Methods

Autoregressive Integrated Moving Average (ARIMA) models remain among the most widely applied statistical approaches for forecasting time series that exhibit trend and autocorrelation, and continue to form the theoretical foundation for univariate export and demand forecasting studies published in recent years (Fattah et al., 2018). An ARIMA model is defined by three parameters, denoted ARIMA(p, d, q), each corresponding to a distinct structural component of the series (Fatteh et al., 2018).

2.4.1 Stationarity and differencing

A core requirement of ARIMA modelling is that the series being modelled must be stationary that is, its statistical properties (mean, variance, and autocorrelation) must not change over time. Malaysia's E&E export series, like most economic time series, exhibits a clear long-run upward trend and is therefore non-stationary in its raw form. This non-stationarity must be formally verified before model fitting proceeds, using a unit-root test such as the Augmented Dickey-Fuller (ADF) test, which tests the null hypothesis that a unit root is present (i.e., the series is non-stationary). Where a series is found to be non-stationary, differencing subtracting each observation from the previous one is applied to stabilise the mean and remove trend; the number of differencing operations required is denoted by the d parameter. Both the ADF

verification step and the sufficiency of first-order differencing for a persistently trending economic series have recently been demonstrated in applications to demand and commodity-price forecasting (Fatteh et al., 2018; Ohyver and Pudjihastuti, 2018).

2.4.2 Autoregressive and moving-average components

Once stationarity is achieved, the p and q parameters capture the series' remaining autocorrelation structure. The autoregressive (AR) component models the current value of the series as a linear function of its own p previous values, capturing the persistence or momentum inherent in the data. The moving-average (MA) component models the current value as a function of q previous forecast errors, capturing short-term shocks that take several periods to fully dissipate. A low-order combination of both components as few as one autoregressive and one moving-average term has been shown sufficient to capture this structure in monthly demand data without requiring higher-order specifications (Fatteh et al., 2018).

2.4.3 Model identification and order selection

The classical Box-Jenkins methodology identifies appropriate values of p and q by examining the autocorrelation function (ACF) and partial autocorrelation function (PACF) of the (differenced) series. In practice, this visual identification is supplemented and increasingly superseded by comparing multiple candidate model specifications using information criteria such as the Akaike Information Criterion (AIC) or the Bayesian Information Criterion (BIC), which balance model fit against model complexity. This two-stage process ACF/PACF narrowing followed by formal comparison across several candidate specifications using AIC, BIC, and related fit statistics is the approach this study follows in Section 5.4 when selecting among six candidate ARIMA($p,1,q$) specifications for Malaysia's E&E export series (Fatteh et al., 2018; Ohyver and Pudjihastuti, 2018).

2.4.4 Seasonality

Where a series exhibits recurring patterns at fixed intervals, the seasonal variant, SARIMA(p, d, q)(P, D, Q)_s, extends the basic ARIMA framework with additional seasonal terms operating at the seasonal period *s*. Incorporating these seasonal AR, differencing, and MA terms has been shown to materially improve forecast accuracy over a non-seasonal specification when the underlying series contains a genuine recurring cycle, particularly when benchmarked against alternative seasonal methods such as Holt-Winters exponential smoothing. This is directly relevant to the present study: Sheng and Ismail applied a SARIMA model to Malaysia's E&E sales data and found it captured the series' seasonal and trend dynamics effectively, though their analysis was constrained to quarterly rather than monthly resolution — a limitation this study addresses by retaining monthly data throughout (Pongdatu and Putra, 2018; Sheng and Ismail, 2025).

2.4.5 Diagnostic checking

After a candidate model is fitted, its adequacy is verified through residual diagnostic checks: residuals should resemble white noise, with no remaining significant autocorrelation, assessed via the Ljung-Box test. This diagnostic step is standard practice in recent applied ARIMA export studies, where a fitted export series is only accepted once its standardised residuals and residual ACF are confirmed to show no remaining structure and residual p-values remain above the significance threshold across lags (Islam, 2020).

2.4.6 Empirical relevance to export forecasting

Beyond Sheng and Ismail's SARIMA application, ARIMA-based approaches have demonstrated value in export-forecasting contexts elsewhere in the recent literature. Islam applied the Box-Jenkins ARIMA methodology to Bangladesh's tea production, internal consumption, and export series, identifying an ARIMA(0,1,1) specification for exported tea after confirming the series was non-stationary and became stationary under first differencing, and validating the selected model through

standardised residual and autocorrelation diagnostics before forecasting. Qiu similarly evaluated multiple modelling approaches — including time-series methods incorporating the U.S. Industrial Production Index — for forecasting China's export value to the United States, finding satisfactory forecasting performance and lending support to the use of external industrial-demand indicators alongside a series' own historical structure (Islam, 2020; Qiu, 2022; Sheng and Ismail, 2025). This body of evidence, combined with ARIMA's capacity to model a series purely from its own historical structure without requiring an external predictor set that must itself be forecast or assumed at future time points, provides the literature-based justification for selecting ARIMA as the primary forecasting method in this study.

2.5 Research Gap

Despite the established relevance of exchange rate, inflation, and external industrial demand to Malaysia's E&E and manufacturing export performance (Wong, 2008; Qiu, 2022; Baharin, Shahperi and Guangqin, 2023; Hakim, 2024), and despite the demonstrated effectiveness of SARIMA for Malaysia's E&E sector specifically study written by Sheng and Ismail, no existing study has developed and rigorously validated an ARIMA model for Malaysia's monthly E&E export series. It studies SARIMA application the closest related work is constrained to quarterly sales data rather than monthly export values, reducing both the resolution and the number of observations available for robust model identification (Sheng and Ismail, 2025). Existing Malaysia-specific econometric studies of E&E and manufacturing exports (Wong, 2008; Muhammad and Yaacob, 2009; Hakim, 2024), meanwhile, focus on estimating long-run cointegrating relationships between exports and macroeconomic conditions rather than producing operational short-term forecasts.

This leaves a clear gap: a properly identified, diagnostically validated, monthly-resolution ARIMA forecasting model for Malaysia's E&E exports, evaluated across the one-, two-, and three-month horizons relevant to the short-term decision-making needs of policymakers, trade analysts, and E&E industry stakeholders. This study addresses that gap directly, and additionally provides supplementary empirical evidence presented in Appendix on how multivariate machine-learning approaches

(Random Forest, SVR, XGBoost) perform against this ARIMA benchmark, informed by the non-linear relationships documented in the broader machine-learning literature (Sengupta, Pratap and Pawar, 2025).

CHAPTER 3

RESEARCH METHODOLOGY

3.1 Introduction

In this chapter, a methodology for developing an ARIMA-based forecasting system for Malaysia's Electrical and Electronics (E&E) exports is described. The methodology follows the CRISP-DM (Cross-Industry Standard Process for Data Mining) framework, structured across six phases: Business Understanding, Data Understanding, Data Preparation, Modelling, Evaluation and Deployment. ARIMA is developed as the primary forecasting method, using only the historical E&E export series itself, following a structured model-identification process comprising stationarity testing, differencing, order selection, model fitting, and diagnostic checking. The historical relationship between E&E exports and three selected macroeconomic indicators inflation, the MYR/USD exchange rate, and the U.S. Industrial Production Index is examined descriptively to support RQ1, but these indicators are not used as ARIMA model inputs, since ARIMA is a univariate method. As supplementary work, Random Forest, Support Vector Regression (SVR), and XGBoost were also developed and evaluated against the ARIMA model; their methodology, implementation, and results are presented in Appendix A.

3.2 Research Design

This study uses a quantitative, empirical and data-driven research design to develop an ARIMA-based forecasting system for Electrical and Electronics (E&E) export values in Malaysia. The study employs a time-series forecasting method using monthly data from January 1990 to March 2025. Monthly frequency permits the study to reflect changes in the performance of E&E exports, seasonal trends, and short-term variation across the series. The forecasting framework is intended to produce one-, two- and three-month-ahead forecasts.

Monthly observations are used instead of quarterly or annual observations to preserve short-term information that would otherwise be averaged away at a higher level of aggregation, and because the forecast horizons of interest are themselves short-term. The dataset contains 423 monthly observations from January 1990 to March 2025, offering sufficient historical depth to observe multiple full economic cycles — including periods of expansion, the 2008 Global Financial Crisis, and the COVID-19 pandemic — which is necessary for a robust ARIMA model identification process rather than one that reflects only a single economic regime.

The study examines the historical structure of Malaysia's E&E export series as the sole basis for ARIMA forecasting, since ARIMA forecasts each future value from the series' own past observations and forecast errors. Separately, the study examines the descriptive relationship between E&E exports and three selected economic indicators — inflation, the MYR/USD exchange rate, and the U.S. Industrial Production Index — to address RQ1 and provide broader contextual understanding of the series' economic environment; these indicators are not incorporated into the ARIMA model itself.

3.3 CRISP-DM Framework

The primary method for developing the ARIMA-based forecasting system in this study is based on the Cross-Industry Standard Process for Data Mining (CRISP-DM). The CRISP-DM process is an iterative and structured research process consisting of six phases: Business Understanding, Data Understanding, Data Preparation, Modelling, Evaluation and Deployment.

CRISP-DM was chosen over other process frameworks, such as SEMMA (Sample, Explore, Modify, Model, Assess) and KDD (Knowledge Discovery in Databases), because it was explicitly designed around iterative feedback between business goals and technical development, an approach not explicitly built into these alternatives. This is appropriate for a forecasting study such as this one, where diagnostic checking results (Chapter 5) may indicate that an ARIMA specification should be revisited. CRISP-DM is also the most widely adopted and vendor-neutral

standard for structuring applied data-science projects, providing a transparent and replicable structure for reporting this study's methodology (Chapman *et al.*, 2000).

In this study, the Business Understanding phase defines the need to forecast Malaysia's E&E export values over a short-term horizon using ARIMA, and establishes the research objectives and success criteria. The Data Understanding phase involves collecting and examining the monthly E&E export series, alongside the three macroeconomic indicators used for descriptive analysis. The Data Preparation phase covers data cleaning, integration, and specific to ARIMA stationarity testing and differencing of the export series. During the Modelling phase, the ARIMA model is identified (order selection via AIC) and fitted; as supplementary work, Random Forest, SVR, and XGBoost are also developed (Appendix A). The Evaluation phase assesses ARIMA's forecasting performance using MSE, RMSE, MAE and R^2 across the selected forecasting horizons, including validation against genuinely unseen data. Finally, the Deployment phase presents the forecasting results through an interactive Power BI dashboard.

CRISP-DM is iterative rather than strictly sequential. For example, the diagnostic checking results presented in Chapter 5 — which revealed residual autocorrelation in the fitted ARIMA model — motivate revisiting the model specification (e.g., considering a seasonal ARIMA extension), illustrating exactly the kind of feedback between later and earlier stages that CRISP-DM is designed to accommodate.

3.4 Research Methodology Workflow

In this study, the overall research process follows the CRISP-DM framework and covers the stages from business understanding and data collection to ARIMA model identification, fitting, diagnostic checking, evaluation, and dashboard deployment.

The workflow begins with Business Understanding, where the forecasting problem, research objectives and success criteria are defined. This is followed by Data

Understanding, involving collection of the monthly E&E export series and the three macroeconomic indicators. Data Preparation involves cleaning and integrating the datasets into a consistent monthly structure, followed by stationarity testing and differencing of the export series specifically for ARIMA modelling. The Modelling stage involves identifying the ARIMA order via AIC comparison, fitting the model, and conducting residual diagnostic checks (Ljung-Box test). The model is evaluated for one-, two-, and three-month-ahead forecasting using MSE, RMSE, MAE and R^2 , and validated against out-of-sample data. Finally, the Deployment stage presents the historical trends, ARIMA model diagnostics, and forecast results through an interactive Power BI dashboard.

As supplementary work, Random Forest, SVR, and XGBoost are developed in parallel using the engineered lag, rolling, and macroeconomic-predictor features described in Appendix A, providing a secondary point of comparison against the ARIMA model.

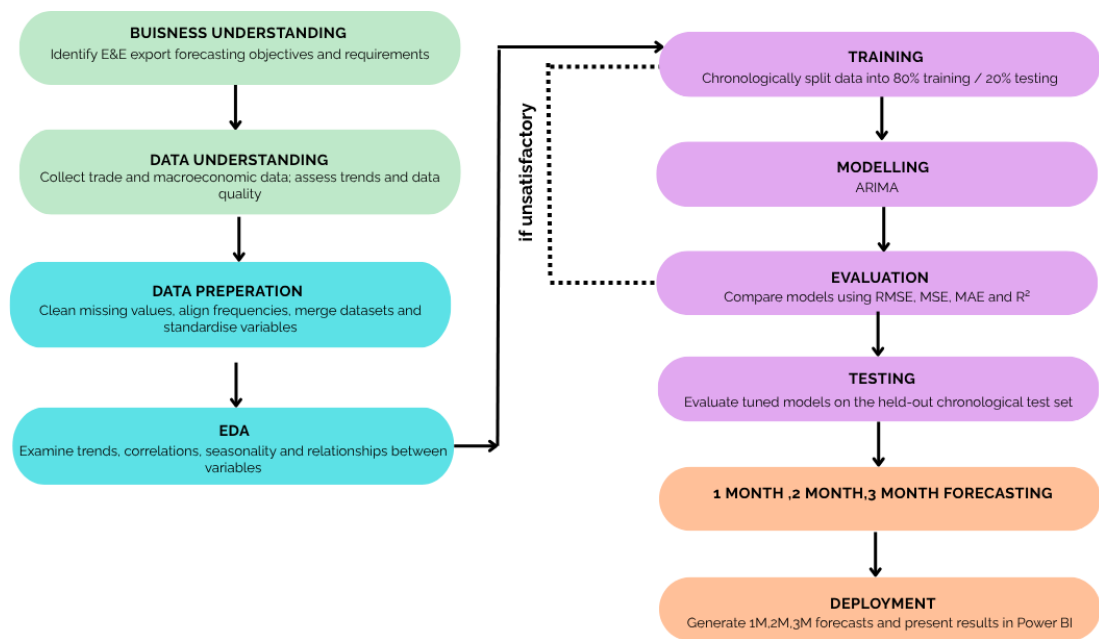


Figure 3:1 Research Workflow Diagram

3.4.1 Business Understanding

3.4.1.1 Project Objectives

The project pursues three primary objectives:

- (a) To analyse the patterns, trends and relationships between Malaysia's monthly E&E export values, historical time-series characteristics and the selected macroeconomic indicators.
- (b) To develop, validate and evaluate an ARIMA time-series model for forecasting Malaysia's monthly E&E export values at one-, two- and three-month-ahead horizons.
- (c) To develop an interactive Power BI dashboard for presenting historical E&E export trends, model performance and one-, two- and three-month-ahead forecast results to support evidence-based decision-making.

The stakeholders identified in Section 1.4 are interested in near-term forecasts, as their planning and decision-making processes have short time horizons: policymakers and government agencies rely on short-term forecasts to calibrate trade responses; E&E exporters are interested in one-, two- and three-month forecasts for planning production and inventory; financial institutions are interested in short-term forecasts for making lending and risk decisions. As supplementary work, machine-learning models (Random Forest, SVR, XGBoost), which incorporate macroeconomic indicators and engineered time-series features, are also developed and compared against ARIMA to assess whether their additional complexity offers a forecasting advantage over the univariate statistical model for these same short time horizons (Appendix A).

3.4.1.2 Success Criteria

The success of this study is primarily assessed based on the forecasting performance of the ARIMA model for Malaysia's E&E exports. Model performance is evaluated using Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R^2 . Lower MAE, MSE and RMSE values indicate lower prediction errors, while a higher R^2 value indicates a greater proportion of variation in the target variable being explained by the model.

These four metrics were selected because together they provide complementary information about forecasting error: MAE gives an easily interpretable average error magnitude, MSE and RMSE penalise larger errors more heavily (important given the practical cost of large planning errors), and R^2 indicates the proportion of variation in E&E export values explained by the model.

The model's performance will be assessed based primarily on RMSE, with MAE, MSE and R^2 used as supporting measures. Evaluation will be performed across the three forecasting horizons, alongside out-of-sample validation against genuinely unseen recent data (Section 5.8). The final model should demonstrate acceptable forecasting performance, while residual diagnostic results are examined to identify whether important autocorrelation remains unexplained and produce forecasts that can be effectively communicated through the proposed Power BI dashboard. The same evaluation framework is applied to the supplementary machine-learning comparison (Appendix A) to allow a consistent basis for comparison.

3.4.1.3 Project Planning

This section outlines the overall execution plan for the project, covering the methodological framework, resources, tools, and timeline required to achieve the research objectives stated in Section 3.4.1.1. The project follows the CRISP-DM (Cross-Industry Standard Process for Data Mining) methodology, which provides a structured sequence of phases — from business and data understanding through to data preparation, modelling, evaluation, and deployment — ensuring that each stage of the

ARIMA model development is systematically planned and tracked. The plan further identifies the data, computing, and human resources needed, the software tools used for implementation, and an 8-week timeline mapping each project phase to its corresponding research objective. Table 3:1 summarises this project execution plan.

Table 3:1 Project Execution Plan

Category	Description
Project Plan	The project follows the CRISP-DM methodology, beginning with problem definition and business understanding, followed by data collection, data understanding, data preparation, stationarity testing, ARIMA model identification and fitting, diagnostic checking, evaluation, and deployment through an interactive forecasting dashboard.
Data Resources	Historical monthly Malaysia E&E export data, used as the sole input to the ARIMA model, and three descriptive macroeconomic indicators — CPI/inflation, MYR/USD exchange rate and the U.S. Industrial Production Index — sourced from the Ministry of Economy Malaysia, DOSM, and FRED respectively.
Computing Resources	A typical personal computer with sufficient processing capacity for data preprocessing, ARIMA model fitting, diagnostic checking, and (for the supplementary comparison) machine-learning model training and tuning.
Human Resources	The project is implemented by the researcher with academic supervision and regular monitoring and guidance throughout the project development process.
Tools and Software	Python is the primary programming language. Data preprocessing uses pandas and NumPy; statsmodels is used for ARIMA modelling, stationarity testing, and diagnostic checking. As supplementary work, scikit-learn and XGBoost are used for the machine-learning comparison (Appendix A). Power BI is used to develop the dashboard.
Timeline	The project is divided into 8 weeks: Weeks 1–2: Data Collection and Understanding; Weeks 3–4: Data Cleaning, Stationarity Testing and Differencing; Weeks 5–6: ARIMA Order Selection, Fitting and Diagnostic Checking; Weeks 7–8: Evaluation, Deployment and Final Report.

This timeline maps directly onto the CRISP-DM phases and research objectives: Weeks 1–2 correspond to Business Understanding and Data Understanding and support RO1; Weeks 3–4 correspond to Data Preparation and support RO1; Weeks 5–6 correspond to Modelling and support RO2; and Weeks 7–8 correspond to Evaluation and Deployment and support RO2 and RO3.

3.4.2 Data Understanding

3.4.2.1 Data Sources and Acquisition

The data used in this study were collected from official and established data sources to provide consistent and credible information for forecasting Malaysia's Electrical and Electronics (E&E) exports. The study uses monthly observations covering the period from January 1990 to March 2025, resulting in 423 monthly observations. The long observation period provides sufficient historical variation for examining changes in E&E export performance across different economic conditions and supports the development and evaluation of the ARIMA forecasting model.

The data used in this study were collected from four official and established sources, summarised in the table below.

Table 3:2 Data Sources

Variable	Description	Source Organisation	Access Method	Study Period	Access Date	Frequency	Unit	Role
E&E Exports	Malaysia's E&E export value	Ministry of Economy Malaysia – External Trade Statistics	Public portal download	Jan 1990 – Mar 2025	March 2025	Monthly	RM million	Target
CPI Inflation	Malaysia's inflation rate	Department of Statistics Malaysia (DOSM)	Public portal download (OpenDOSM)	Jan 1990 – Mar 2025	March 2025	Monthly	% change	Descriptive variable / supplementary ML predictor
MYR/USD Exchange Rate	MYR to USD exchange rate	Federal Reserve Bank of St. Louis (FRED)	Public download via FRED	Jan 1990 – Mar 2025	March 2025	Monthly	MYR per USD	Descriptive variable / supplementary ML predictor
U.S. Industrial Production Index	U.S. industrial output index	Federal Reserve Bank of St. Louis (FRED)	Public download via FRED	Jan 1990 – Mar 2025	March 2025	Monthly	Index (2017=100)	Descriptive variable / supplementary ML predictor

All four datasets were sourced from official government or central-bank-affiliated institutions rather than third-party aggregators, which supports their reliability and consistency for academic research. The Ministry of Economy Malaysia and DOSM are the official Malaysian government sources for trade and price statistics respectively, while FRED is a widely used and academically recognised repository maintained by the Federal Reserve Bank of St. Louis for macroeconomic time-series data. All datasets were accessed and downloaded in March 2025 to align with the end of the study's observation period.

The forecasting target is Malaysia's monthly E&E export value, obtained from the Ministry of Economy Malaysia's External Trade Statistics portal. The predictor variables consist of CPI/inflation, the MYR/USD exchange rate, and the U.S. Industrial Production Index. CPI/inflation data are obtained from the Department of

Statistics Malaysia (DOSM), while the exchange rate and U.S. Industrial Production Index are obtained from the Federal Reserve Bank of St. Louis (FRED).

Monthly frequency is adopted because the study focuses on short-term forecasting of E&E exports up to three months ahead. Using monthly observations allows the model to capture more frequent changes in export performance and movements in the selected economic indicators compared with an annual dataset. The monthly frequency therefore provides an appropriate temporal resolution for the forecasting objective.

The datasets were standardised before integration to ensure consistency in their date formats, units and variable names. The Date field was used as the common time index for combining the datasets. After standardisation, the datasets were integrated into a single multivariate time-series dataset containing the E&E export target and the selected predictor variables. The resulting dataset provides a consistent monthly structure for subsequent exploratory analysis and ARIMA modelling.

The E&E export series serves as the sole input to the ARIMA model developed in this study. The three macroeconomic indicators — CPI/inflation, the MYR/USD exchange rate, and the U.S. Industrial Production Index — are examined descriptively in Chapter 4 to address RQ1, and are used as predictor variables only in the supplementary machine-learning comparison presented in Appendix A.

3.4.3 Data Preparation

Data preparation was conducted to transform the collected monthly datasets into a consistent format suitable for machine-learning-based forecasting. The preparation process included data selection, data cleaning, data integration and formatting, followed by exploratory analysis and preparation of the E&E export series for ARIMA modelling. These steps were performed to ensure that the E&E export target and selected economic predictors were appropriately structured while preserving the chronological nature of the time-series data.

3.4.3.1 Data Selection

The E&E export series was selected as the sole variable for ARIMA model development, consistent with ARIMA's univariate design: the model forecasts future export values directly from the series' own historical structure, without requiring external predictor data at forecast time.

Three additional variables inflation, the MYR/USD exchange rate, and the U.S. Industrial Production Index were selected for descriptive analysis addressing RQ1, based on their relevance to Malaysia's E&E export performance as established in the literature (Chapter 2) and their availability throughout the study period. Inflation was included because changes in domestic price levels can affect production costs and export price competitiveness. The MYR/USD exchange rate was selected because exchange-rate movements can affect the relative price competitiveness of Malaysian exports. The U.S. Industrial Production Index was selected as an external indicator of industrial activity and demand conditions relevant to Malaysia's export-oriented E&E sector.

The exclusion of other potentially relevant variables GDP (quarterly frequency only) and a dedicated semiconductor-demand index (no consistent monthly series available) was a deliberate methodological choice, discussed further in Section 1.6.

As supplementary work, these three indicators are also incorporated as predictor variables in the machine-learning comparison presented in Appendix A, where their relevance is further supported by the literature reviewed in Chapter 2, Section 2.5.

3.4.3.2 Data Cleaning

The source datasets were cleaned to improve consistency and ensure that the observations were suitable for subsequent integration and modelling. Non-relevant rows, including subtotal rows, metadata and empty records, were removed, while only observations relevant to the monthly E&E export forecasting analysis were retained.

Missing-value and duplicate checks were performed on each of the four analytical variables (E&E Exports, CPI Inflation, Exchange Rate and U.S. IPI) across all 423 monthly observations. No missing values and no duplicate records were identified in any of the four variables (detailed results are presented in Table 4:2, Chapter 4), and therefore no imputation or deduplication was required. Each variable was also visually inspected for outliers (boxplots and time-series plots) and the interquartile range (IQR) method was used. High and low points of the E&E export and inflation series could be attributed to genuine and identifiable economic events such as the 2008 Global Financial Crisis, the 2022 inflation surge and the COVID-19 pandemic, not to data entry errors. Such observations were therefore not deleted or capped from the data, but were instead kept so as not to lose the true historical variability that is important for the forecasting problem.

The cleaned datasets were then prepared for integration using the common monthly date index. The resulting data were checked for inconsistencies in data types, duplicate observations and chronological ordering before proceeding to data integration and feature construction.

3.4.3.3 Data Integration and Formatting

Before integration, each of the four source datasets contained 423 monthly observations. A temporary Date_YM key in YYYY-MM format was created for each dataset to ensure that observations were merged on the same calendar month rather than exact date, since the original datasets used differing date conventions and reporting frequencies. The E&E export dataset was used as the base table and progressively left-joined with CPI inflation, exchange rate and U.S. IPI data on this common monthly key. After integration, the temporary key was removed, the date field was converted to a proper datetime index, and the observations were sorted chronologically. The resulting integrated dataset retained all 423 monthly observations with no rows lost or duplicated during merging, and post-integration checks confirmed zero missing values and zero duplicate records across the combined dataset (Table 4:3, Chapter 4). The integrated dataset was subsequently prepared for feature construction and model development. The chronological ordering was also maintained when

separating the data into training and testing periods to avoid using future observations to predict earlier periods.

3.4.3.4 Data Exploration

Exploratory Data Analysis (EDA) was conducted to understand the structure, trends, patterns and relationships within the integrated monthly dataset before ARIMA model development. The analysis focused primarily on the historical behaviour of Malaysia's E&E export series, while the selected macroeconomic indicators were examined to provide additional contextual understanding.

Time-series visualisations were used to examine the long-term trend, fluctuations and unusual movements in Malaysia's E&E export values. Major economic events, including the 2008–09 Global Financial Crisis and the COVID-19 period, were considered when interpreting substantial changes in export performance. These events were retained as part of the historical series because they represent genuine economic conditions rather than data errors.

Seasonal analysis was conducted to determine whether recurring monthly patterns were present in the E&E export series. Classical seasonal decomposition with a 12-month period was used to separate the series into trend, seasonal and residual components. Monthly patterns were also examined to identify whether particular months consistently showed higher or lower export values.

Descriptive statistics, including the mean, median, standard deviation, minimum and maximum values, were calculated to characterise the distribution and variability of the E&E export series and the selected macroeconomic indicators.

Correlation analysis was conducted to examine the direction and strength of the linear relationships between E&E exports and CPI inflation, the MYR/USD exchange rate and the U.S. Industrial Production Index. The correlation results were treated as exploratory and contextual evidence rather than proof of causality or predictive significance

For ARIMA modelling, stationarity and autocorrelation were also examined. The Augmented Dickey-Fuller (ADF) test was applied to assess whether the E&E export series was stationary. The Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) were examined to understand the dependence structure of the series and provide preliminary guidance for ARIMA order selection.

The detailed EDA results are presented in Chapter 4. The findings from the EDA were subsequently used to support the ARIMA model identification, particularly the decisions regarding differencing and candidate model orders.study.

3.4.4 Modelling

The modelling stage focuses primarily on developing an Autoregressive Integrated Moving Average (ARIMA) model for forecasting Malaysia's monthly E&E export values. ARIMA was selected as the main forecasting method because it can model the historical structure of a time series using its own past observations and forecast errors.

The ARIMA modelling process follows the Box-Jenkins methodology. The process begins by assessing the stationarity of the E&E export series using the Augmented Dickey-Fuller (ADF) test. If the series is non-stationary, differencing is applied until stationarity is achieved. The ACF and PACF are then examined to provide preliminary guidance for selecting suitable autoregressive (p) and moving-average (q) orders.

Several candidate ARIMA specifications are subsequently fitted to the training data. The candidate models are compared using the Akaike Information Criterion (AIC), with the model having the lowest AIC considered the preferred specification, provided that its diagnostic results are satisfactory.

After model selection, the selected ARIMA model is fitted to the training data and its residuals are examined. The Ljung-Box test is used to determine whether significant autocorrelation remains in the residuals. A suitable model should produce

residuals that behave approximately as white noise, indicating that the important time-series structure has been captured.

The selected ARIMA model is then used to generate one-, two- and three-month-ahead forecasts. Forecast performance is evaluated using the predefined evaluation metrics and subsequently validated using genuinely unseen observations.

Random Forest, Support Vector Regression (SVR) and XGBoost are retained only as supplementary comparison models. Their detailed methodology and implementation are presented separately in Appendix A and do not form the primary modelling methodology of this study.

3.4.4.1 Model Selection

ARIMA was selected as the primary forecasting method for this study based on the literature-supported justification developed in Chapter 2, Section 2.4: its capacity to model a series purely from its own historical structure, without requiring external predictor data that must itself be forecast or assumed at future time points, combined with demonstrated effectiveness in comparable export-forecasting contexts (Sheng and Ismail, 2025; Qiu, 2022).

As supplementary work, three machine-learning models — Random Forest, SVR, and XGBoost — were also developed to evaluate whether incorporating macroeconomic indicators and engineered time-series features via multivariate approaches offers a forecasting advantage over the univariate ARIMA model. These models were selected for their methodological diversity (bagging, kernel-based, and boosting approaches respectively) and their suitability for the dataset's moderate size, as detailed in Appendix A.

Deep-learning models such as LSTM were not considered for either the primary or supplementary analysis, given the relatively limited number of monthly observations available.

3.4.4.2 Data Splitting Strategy

Since this study is of a time-series forecasting nature, a random train-test split was not employed as this might include information from future periods in the training set. Instead, a chronological 80:20 split was applied to the development sample, with the earliest 80% used for initial model development and the most recent 20% retained for historical walk-forward testing. 80% of observations were used for model training and validation, and 20% for the most recent observations were retained as an unseen test set.

This chronological splitting approach maintains the temporal sequence of the observations and more closely mimics a real-world forecasting problem, in which future values are to be forecast based upon the information from the previous time periods. The same chronological evaluation period was maintained for Random Forest, SVR and XGBoost to ensure a fair comparison between the models.

Concretely, the training partition covers the earliest 326 monthly observations, and the testing partition covers the most recent 82 monthly observations, held out entirely from training. An 80:20 split was selected as it is a widely used convention in applied forecasting studies that balances providing sufficient historical data for the models against retaining a test period long enough (82 months, approximately seven years) to evaluate performance across varied economic conditions rather than a narrow or unrepresentative window.

The full dataset contained 423 monthly observations from January 1990 to March 2025. For historical model development and walk-forward testing, observations from January 1990 to December 2023 were used, providing 408 observations. This development sample was divided chronologically into an initial training partition comprising the earliest 326 observations and a historical test partition comprising the subsequent 82 observations. The observations from January to December 2024 were subsequently incorporated when retraining the final operational model, while January to March 2025 remained excluded from model fitting and was used for final out-of-sample validation.

Table 3:3 Data Splitting Strategy

Partition	Observations	Period	Purpose
Initial training	326	Earliest observations in Jan 1990–Dec 2023 development sample	Stationarity testing, order selection and initial model fitting
Historical test	82	Final observations in Jan 1990–Dec 2023 development sample	Walk-forward forecasting evaluation
Model-updating period	12	Jan–Dec 2024	Incorporated before fitting the final operational model
Final validation	3	Jan–Mar 2025	Out-of-sample validation only

3.4.4.3 Model Training Process

The ARIMA model was identified and trained following the Box-Jenkins methodology described in Chapter 2, Section 2.4. First, the Augmented Dickey-Fuller (ADF) test was applied to confirm non-stationarity of the raw export series and to verify stationarity after first-order differencing (Section 5.3). Second, six candidate ARIMA(p,1,q) specifications were fitted to the training data and compared using the Akaike Information Criterion (AIC), with the lowest-AIC specification — ARIMA(2,1,2) — selected as the final model order (Section 5.4). Third, the selected model was fitted and its residuals checked for remaining autocorrelation using the Ljung-Box test (Section 5.5). Finally, forecasts were generated using a walk-forward (expanding-window) procedure, in which the model is refit at each point in the test period as new actual observations become available, rather than forecasting the entire test period from a single static origin (Section 5.6).

At each forecast origin, the model was fitted using all observations available up to that origin and used to generate forecasts up to three months ahead. After the forecasts had been stored, only the next actual observation was added to the expanding training history. This produced 82 one-month-ahead forecasts, 81 two-month-ahead forecasts and 80 three-month-ahead forecasts.

Table 3:4 ARIMA Model Identification and Training Procedure

Stage	Procedure
Stationarity testing	Augmented Dickey-Fuller (ADF) test applied to the raw and first-differenced E&E export series
Order selection	Six candidate ARIMA(p,1,q) specifications compared via AIC; lowest-AIC order selected
Model fitting	Selected ARIMA order fitted on the training partition (326 observations)
Diagnostic checking	Ljung-Box test applied to model residuals to assess remaining autocorrelation
Forecasting	Walk-forward (expanding-window) procedure applied across the 82-observation test period, refitting the model at each step

As supplementary work, Random Forest, SVR, and XGBoost were also trained and tuned using five-fold time-series cross-validation, with hyperparameter search ranges and procedures detailed in Appendix A.

3.4.4.4 Model Evaluation Metrics

The forecasting performance of the selected ARIMA model was evaluated using Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2). These metrics were calculated for the one-, two-, and three-month-ahead forecasts to assess the accuracy of the model at different forecasting horizons.

The evaluation results were used to assess the forecasting accuracy of the ARIMA model across the different forecasting horizons.

Table 3:5 Performance Evaluation Criteria

Metric	What it measures	Interpretation	Equation
MAE	Average absolute difference between actual and forecast values.	Lower values indicate smaller average forecasting errors.	$\frac{1}{n} \sum_{i=1}^n y_i - \hat{y}_i $
MSE	Average squared difference between actual and forecast values.	Lower values indicate smaller errors, while larger errors receive greater penalty.	$\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$
RMSE	Square root of MSE, expressed in the same unit as the E&E export target.	Lower values indicate better forecasting accuracy and provide an interpretable measure of forecast error.	$\sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$
R ²	R ² is used here as an out-of-sample comparative goodness-of-fit measure. It compares the forecast errors against a mean-based reference calculated over the corresponding evaluation observations. A negative value indicates that the forecasts perform worse than that reference.	Higher values indicate better explanatory performance, with values closer to 1 indicating stronger fit.	$\sqrt{1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}}$

3.4.5 Evaluation

The evaluation stage assesses the forecasting performance and practical relevance of the developed models. The evaluation considers both quantitative forecasting accuracy and the ability of the models to represent the overall movement of Malaysia's E&E export values. The results from the evaluation stage are used to identify the most suitable model for generating the final three-month-ahead forecasts.

3.4.5.1 Technical Performance Evaluation

The technical performance of the ARIMA model will be evaluated using MAE, MSE, RMSE and R^2 across the one-, two-, and three-month-ahead forecasting horizons, with greater emphasis placed on RMSE where evaluation metrics disagree. In addition to the walk-forward test-period evaluation, the final model's performance will be validated against genuinely unseen out-of-sample data (January–March 2025), providing a real-world check on forecasting reliability beyond the historical test period (Section 5.8). As supplementary work, the same evaluation framework is applied to Random Forest, SVR, and XGBoost (Appendix A), enabling a comparison of the univariate ARIMA model against multivariate machine-learning alternatives.

3.4.5.2 Business Relevance Assessment

The business relevance of the forecasting models will be assessed based on their potential usefulness in monitoring and anticipating Malaysia's E&E export performance. In addition to the quantitative evaluation metrics, actual and predicted E&E export values will be compared using time-series visualisations to assess whether the models can reasonably capture the overall trends and fluctuations in export performance.

The selected model will then be used to generate the three-month-ahead E&E export forecasts based on the most recent available information. The forecast results will subsequently be presented through the Power BI dashboard to support the

interpretation and communication of Malaysia's expected short-term E&E export performance.

As no formal stakeholder consultation or expert evaluation was conducted as part of this study, the following should be understood as an interpretation of the forecasts' potential relevance to each stakeholder group rather than a formally validated business assessment. For government agencies and policymakers, the forecasts may help flag anticipated declines or surges in E&E export performance early enough to inform policy discussion. For E&E exporters and businesses, the one-to-three-month horizon aligns with typical production and inventory planning cycles, potentially supporting more informed short-term operational decisions. For financial institutions and market analysts, the forecasts and associated model-performance information may provide supplementary context when assessing sector-level risk and trade exposure, alongside their own analysis.

3.4.5.3 Process Review and Iteration

Model development followed an iterative process in which the ARIMA order selection and diagnostic checking results were reviewed to confirm the adequacy of the selected specification. Where diagnostic checks revealed remaining residual structure (Section 5.5), this was documented as a limitation informing future recommendations (Section 7.8) rather than triggering a full model respecification within the scope of this study. The final model was retrained on the complete available historical series before generating out-of-sample forecasts.

3.4.5.4 Determining Next Steps

Based on the model evaluation and diagnostic checking process, the ARIMA (2,1,2) model was confirmed as the final forecasting model for this study. Unlike the earlier framing of this study, in which ARIMA served only as a statistical benchmark against which separate machine-learning models were operationally deployed, ARIMA is now the primary forecasting method throughout: the same model used for

test-period evaluation is also the model used to generate operational forecasts and to conduct out-of-sample validation.

To validate the final model against genuinely unseen data, the ARIMA(2,1,2) model was fitted on all observations through December 2024 and used to forecast three steps ahead January, February, and March 2025 with these forecasts compared directly against the actual values available for this period (Section 5.8). This out-of-sample validation provides a real-world check on the model's forecasting reliability, complementing the walk-forward test-period evaluation.

3.4.6 Deployment

This section describes the Deployment phase of the CRISP-DM methodology, the final stage in which the results of the ARIMA modelling process are made accessible and actionable for stakeholders. Deployment in this study comprises three components: the development of an interactive Power BI dashboard to present forecasting results and model performance (Section 3.4.6.1); a monitoring and maintenance strategy to ensure the model's continued reliability over time (Section 3.4.6.2); and the documentation and reporting of the overall project process (Section 3.4.6.3). Together, these components ensure that the forecasting outputs are not only technically sound but also usable, interpretable, and sustainable for the stakeholders identified in Section 1.4.

3.4.6.1 Dashboard Development and Visualisation

The forecasting results were deployed through a single-page Power BI dashboard designed to present Malaysia's E&E export performance and ARIMA forecasting outputs in an integrated and stakeholder-accessible format. The dashboard consolidates (i) a set of KPI cards summarising the selected ARIMA model specification, the latest available actual E&E export value, the latest forecast value, and the out-of-sample validation error metrics; (ii) a time-series line chart displaying historical monthly E&E exports alongside the ARIMA point forecasts and the associated 95% forecast interval for the validation period; (iii) an out-of-sample

validation table comparing January–March 2025 forecasts against actual values, including error and absolute percentage error; and (iv) walk-forward test-period performance visuals reporting RMSE and R^2 across the one-, two- and three-month-ahead horizons.

To avoid misinterpretation, the dashboard explicitly labels the macroeconomic indicators (CPI inflation, exchange rate and U.S. industrial production) as descriptive context only and not as inputs to the univariate ARIMA model. This single-page design supports multiple stakeholder needs: policymakers and businesses can rapidly interpret recent export movement and short-term forecasts, while analysts and researchers can assess model reliability using the embedded evaluation metrics and forecast-interval coverage.

3.4.6.2 Monitoring and Maintenance Strategy

A monitoring and maintenance plan was devised to support the ARIMA model's continued effectiveness after implementation. Given the ever-changing nature of economic conditions, some decline in forecasting performance over time is expected. Model performance would therefore be evaluated periodically according to MAE, MSE, RMSE and R^2 , with retraining considered when substantial deterioration in forecasting performance is observed. Given the residual autocorrelation identified during diagnostic checking (Section 5.5), periodic re-evaluation should also consider whether a seasonal ARIMA (SARIMA) extension would improve forecasting reliability, as discussed in Chapter 7.

3.4.6.3 Documentation and Reporting

The documentation phase integrates the ARIMA methodology, experimentation, and findings into a reproducible framework. The workflow — from data acquisition and cleaning through stationarity testing, order selection, diagnostic checking, evaluation, and deployment — is systematically documented. As supplementary work, the Random Forest, SVR, and XGBoost implementation and results are documented separately in Appendix A. Conclusions are drawn based on the

ARIMA findings, with recommendations regarding seasonal modelling extensions, forecast-interval estimation, and periodic retraining presented in Chapter 7.

3.5 Chapter Summary

This chapter presented the research methodology adopted for developing an ARIMA-based forecasting system for Malaysia's E&E exports, structured according to the six phases of the CRISP-DM framework. The chapter described the research design and rationale for using monthly data over the January 1990 – March 2025 period; the data sources and preprocessing steps used to prepare the E&E export series and three descriptive macroeconomic indicators; the ARIMA model identification process, comprising stationarity testing, order selection via AIC, model fitting, and diagnostic checking; the chronological data-splitting strategy; and the evaluation metrics (MAE, MSE, RMSE and R^2) used to assess forecasting performance, including out-of-sample validation against genuinely unseen data. As supplementary work, the methodology for a machine-learning comparison (Random Forest, SVR, and XGBoost) was also outlined, with full details presented in Appendix A. Finally, the chapter described the deployment of results through an interactive Power BI dashboard designed to present a complete overview of the historical dataset, model diagnostics, and forecast results. The next chapter presents the results of the exploratory data analysis and data-preparation stages described here.

CHAPTER 4

EXPLORATORY DATA ANALYSIS

4.1 Introduction

This chapter presents the results of the exploratory data analysis and data preparation conducted for the forecasting of Malaysia's Electrical and Electronics (E&E) exports. The analysis examines the characteristics, trends, patterns and relationships within the integrated dataset before the development and evaluation of the forecasting models.

The chapter first presents an overview of the original and final datasets, followed by the results of the data quality assessment and preprocessing procedures. Exploratory analysis is then performed to analyse the historical patterns of E&E exports and the selected macroeconomic indicators such as the MYR/USD exchange rate, U.S. Industrial Production Index and inflation. Key economic upheavals such as the Global Financial Crisis of 2008 and the COVID-19 pandemic also have been taken into account.

In addition, the chapter presents the results of feature construction, including lag features, rolling statistics, growth rates and time-based features. The relationships between E&E exports and the selected predictor variables are examined using correlation analysis. Finally, the resulting feature set and feature importance analysis are presented to provide an understanding of the information used by the forecasting models in the subsequent chapters.

4.2 Dataset Overview

This section provides an overview of the dataset used in this study, describing its composition, sources, and structure prior to the detailed exploratory analysis

presented in subsequent sections. The dataset consists of a single target variable — Malaysia's monthly E&E export value — together with three macroeconomic predictor variables selected for their potential relevance to export performance: CPI inflation, the MYR/USD exchange rate, and the U.S. Industrial Production Index. These variables were integrated into a single monthly time series spanning January 1990 to March 2025, resulting in 423 monthly observations. Section 4.2.1 details the specific sources, access methods, and characteristics of each variable.

4.2.1 Data Sources

The dataset used in this study was constructed by integrating monthly Electrical and Electronics (E&E) export data with selected macroeconomic indicators. The data were obtained from official and publicly available sources to ensure the reliability and consistency of the information used for forecasting.

The monthly E&E export data were obtained from the Ministry of Economy Malaysia External Trade Statistics portal. The dataset represents the value of Malaysia's Electrical and Electronics (E&E) exports and serves as the target variable for the forecasting models. The E&E export data covered the period from January 1990 to March 2025.

The macroeconomic variables used were Malaysia's CPI inflation, MYR/USD exchange rate and U.S. Industrial Production Index (IPI). CPI inflation data are from the Department of Statistics Malaysia (DOSM) and the exchange rate MYR/USD and U.S. Industrial Production Index are from the Federal Reserve Bank of St. Louis (FRED). These variables were selected because they represent domestic price conditions, exchange-rate movements and external industrial activity that may be associated with changes in Malaysia's E&E export performance.

The datasets were obtained at monthly frequency and subsequently standardized using a common monthly date index before integration. The resulting integrated dataset provided the basis for the exploratory analysis, feature construction

and provided the basis for exploratory analysis, univariate ARIMA modelling and the supplementary machine-learning comparison.

Table 4:1 Data Sources

Variable	Description	Source Organisation	Access Method	Study Period	Access Date	Frequency	Unit	Role
E&E Exports	Malaysia's E&E export value	Ministry of Economy Malaysia – External Trade Statistics	Public portal download	Jan 1990 – Mar 2025	March 2025	Monthly	RM million	Target
CPI Inflation	Malaysia's inflation rate	Department of Statistics Malaysia (DOSM)	Public portal download (OpenDOSM)	Jan 1990 – Mar 2025	March 2025	Monthly	% change	Predictor
MYR/USD Exchange Rate	MYR to USD exchange rate	Federal Reserve Bank of St. Louis (FRED)	Public download via FRED	Jan 1990 – Mar 2025	March 2025	Monthly	MYR per USD	Predictor
U.S. Industrial Production Index	U.S. industrial output index	Federal Reserve Bank of St. Louis (FRED)	Public download via FRED	Jan 1990 – Mar 2025	March 2025	Monthly	Index (2017=100)	Predictor

A sample of the E&E export dataset in its original form is shown below, illustrating its structure prior to integration:

SITC 3 DIGIT	DESCRI	1990											
		Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
77	Electric al Machin ery, Apparat us And Applian ces, N.E.S. And Electric al Parts Thereof	1,010	1,035	1,249	1,026	1,223	1,273	1,191	1,343	1,197	1,180	1,263	1,177

Figure 4:1 Original E&E Export Dataset

4.3 Dataset Collecting and Loading

The dataset was collected from multiple data sources and imported into the Python environment for subsequent data preprocessing, transformation, integration, and analysis.

```
import pandas as pd

# Import ER.csv
er_df = pd.read_csv('/content/ER.csv')

# Import CPI Inflation.csv
cpi_inflation_df = pd.read_csv('/content/CPI Inflation.csv')

# Import IPI.csv
ipi_df = pd.read_csv('/content/IPI.csv')

# Import E&E Exports.xlsx
ee_exports_df = pd.read_excel('/content/E&E Exports.xlsx')

print("Files imported successfully!")

Files imported successfully!
```

Figure 4:2 Importing files

4.3.1 Initial Inspection for Each Variables

An initial inspection was conducted for each variable to assess the structure, data type, time coverage, frequency, and completeness of the collected datasets before further preprocessing. The inspection involved examining the date column and recorded values for the E&E export series and the selected macroeconomic variables, namely inflation, exchange rate, and U.S. Industrial Production Index. The number of observations, and consistency of the monthly time-series structure were also

checked. This initial inspection helped any inconsistencies in the data structure that required further preprocessing before the datasets could be integrated.

```
print("\n--- E&E Exports DataFrame Initial Inspection ---")
print(ee_exports_df.head())
print(ee_exports_df.columns)
print("-----")
```

```
--- E&E Exports DataFrame Initial Inspection ---
SITC 3 DIGIT      DESCRIPTIONS \
0      NaN      NaN
1      77.0  Electrical Machinery, Apparatus And Appliances...

      1990  Unnamed: 3  Unnamed: 4  Unnamed: 5  Unnamed: 6  Unnamed: 7 \
0      Jan      Feb      Mar      Apr      May      Jun
1  1010.235516  1034.73632  1249.33347  1025.820057  1223.404799  1273.317912

      Unnamed: 8  Unnamed: 9  ...  Unnamed: 415  Unnamed: 416  Unnamed: 417 \
0      Jul      Aug      ...      Jun      Jul      Aug
1  1190.580015  1342.993456  ...  41083.016724  39357.590921  39863.748126

      Unnamed: 418  Unnamed: 419  Unnamed: 420  Unnamed: 421      2025 \
0      Sep      Oct      Nov      Dec      Jan
1  44006.856395  39226.126364  39795.67259  47002.306925  40494.054767

      Unnamed: 423  Unnamed: 424
0      Feb      Mar
1  36504.760024  45813.712073

[2 rows x 425 columns]
Index(['SITC 3 DIGIT', 'DESCRIPTIONS',      1990, 'Unnamed: 3',
      'Unnamed: 4', 'Unnamed: 5', 'Unnamed: 6', 'Unnamed: 7',
      'Unnamed: 8', 'Unnamed: 9',
      ...,
      'Unnamed: 415', 'Unnamed: 416', 'Unnamed: 417', 'Unnamed: 418',
      'Unnamed: 419', 'Unnamed: 420', 'Unnamed: 421',      2025,
      'Unnamed: 423', 'Unnamed: 424'],
      dtype='object', length=425)
-----
```

Figure 4:3 Initial inspection of the E&E export dataset showing its original wide-format structure and column names

The CPI inflation, exchange rate and U.S. IPI datasets were inspected using the same procedure; each was confirmed to contain a single date column and a single value column at monthly frequency, with no structural inconsistencies requiring correction before transformation.

4.4 Data Preprocessing Results

The table below summarises how the number of observations changed across each preprocessing stage, from the four raw source files through to the final integrated dataset:

Table 4:2 Dataset Observations Before and After Pre-processing

Stage	Observations	Notes
Raw source files (each)	423	Before transformation
After individual transformation	423 each	No observations lost; format-only changes
After merging/integration	423	No observations lost during merge
After missing-value and duplicate checks	423	0 missing values, 0 duplicates found

4.4.1 Data Cleaning Results

Table 4:3 below summarises the cleaning outcome for each of the four datasets. In every case, the cleaning process altered only the format and structure of the data (reshaping, renaming, date standardisation) rather than the underlying numeric values themselves no values were modified, imputed or removed during this stage.

Table 4:3 Data Cleaning Results

Dataset	Observations Before	Observations After	Values Changed?	Format Changed?
E&E Exports	423	423	No	Yes, reshaped from wide to long format
CPI Inflation	423	423	No	Yes, irrelevant columns dropped, date standardised
Exchange Rate	423	423	No	Yes, column names and date field standardised
U.S. IPI	423	423	No	Yes, date field standardised

4.4.1.1 Data Transformation

Data transformation was performed to standardise the variables into a consistent monthly time-series format before integrating them for further analysis and modelling. The transformed dataset contains monthly observations of E&E exports, CPI inflation, exchange rate, and the U.S. Industrial Production Index, with the observation period extending from January 1990 to March 2025. The resulting integrated dataset contains 423 monthly observations. The variables were also renamed to consistent format for easy readability.

The E&E export series was maintained at a monthly frequency, with each observation representing the corresponding monthly export value. The date variable was converted into a standard datetime format to ensure that the observations could be correctly ordered chronologically and used for subsequent time-series feature construction.

```
import pandas as pd
data_start_idx = 2
data_columns_headers = ee_exports_df.columns[data_start_idx:]

years_list = []
current_year = None
for col_header in data_columns_headers:
    # Check if the column header is a year (e.g., '1990', '2025')
    if isinstance(col_header, int) or (isinstance(col_header, str) and col_header.isdigit() and len(col_header) == 4):
        current_year = str(col_header)
        years_list.append(current_year)
years_filled = pd.Series(years_list).ffill().tolist()
months_row_data = ee_exports_df.iloc[0, data_start_idx:]
export_values_data = ee_exports_df.iloc[1, data_start_idx:]
combined_dates = []
for year, month in zip(years_filled, months_row_data):
    if pd.notna(year) and pd.notna(month):
        combined_dates.append(f"{year}-{str(month).strip()}")
    else:
        combined_dates.append(None)
valid_indices = [i for i, date_str in enumerate(combined_dates) if date_str is not None]

ee_exports_transformed_df = pd.DataFrame({
    'Date': [combined_dates[i] for i in valid_indices],
    'E&E Exports': [export_values_data.iloc[i] for i in valid_indices]
})
ee_exports_transformed_df['Date'] = pd.to_datetime(ee_exports_transformed_df['Date'], format='%Y-%b')
ee_exports_transformed_df = ee_exports_transformed_df.sort_values('Date').reset_index(drop=True)
ee_exports_df = ee_exports_transformed_df
```

Figure 4:4 Transformation of the E&E export dataset from the original wide format into a chronological monthly time-series format

The CPI Inflation, Exchange Rate, and U.S. Industrial Production Index were incorporated into the same monthly structure.

```
import pandas as pd
cpi_inflation_transformed_df = cpi_inflation_df.copy()
cpi_inflation_transformed_df = cpi_inflation_transformed_df.rename(columns={'date': 'Date'})
cpi_inflation_transformed_df['Date'] = pd.to_datetime(cpi_inflation_transformed_df['Date'])
if 'division' in cpi_inflation_transformed_df.columns:
    cpi_inflation_transformed_df = cpi_inflation_transformed_df[cpi_inflation_transformed_df['division'] == 'overall']
start_date = '1990-01-01'
end_date = '2025-03-01'
cpi_inflation_transformed_df = cpi_inflation_transformed_df[
    (cpi_inflation_transformed_df['Date'] >= start_date) &
    (cpi_inflation_transformed_df['Date'] <= end_date)
].reset_index(drop=True)
cpi_inflation_transformed_df = cpi_inflation_transformed_df.drop_duplicates(subset=['Date']).reset_index(drop=True)
cpi_inflation_transformed_df = cpi_inflation_transformed_df.rename(columns={'inflation_mom': 'CPI Inflation'})
cpi_inflation_transformed_df = cpi_inflation_transformed_df.drop(columns=['division', 'inflation_yoy'], errors='ignore')
cpi_inflation_df = cpi_inflation_transformed_df
print("Transformed 'CPI Inflation' DataFrame head:")
display(cpi_inflation_df.assign(Date=cpi_inflation_df['Date'].dt.strftime('%Y-%m')).head())
```

Figure 4:5 Transformed CPI inflation dataset after selecting the overall division, retaining the monthly inflation measure and standardising the date variable.

```
import pandas as pd
er_transformed_df = er_df.copy()
er_transformed_df = er_transformed_df.rename(columns={'observation_date': 'Date'})
er_transformed_df = er_transformed_df.rename(columns={'EXMAUS': 'Exchange Rate'})
er_transformed_df['Date'] = pd.to_datetime(er_transformed_df['Date'])
er_df = er_transformed_df
print("Transformed 'Exchange Rate' DataFrame head:")
display(er_df.head())
```

Figure 4:6 Transformed exchange rate dataset after standardising the date and exchange rate variable names.

```
import pandas as pd
ipi_transformed_df = ipi_df.copy()
ipi_transformed_df = ipi_transformed_df.rename(columns={'observation_date': 'Date'})
ipi_transformed_df = ipi_transformed_df.rename(columns={'INDPRO': 'US Industrial Production Index'})
ipi_transformed_df['Date'] = pd.to_datetime(ipi_transformed_df['Date'])
ipi_df = ipi_transformed_df
print("Transformed 'US Industrial Production Index' DataFrame head:")
display(ipi_df.head())
```

Figure 4:7 Transformed U.S. Industrial Production Index dataset after standardising the date and variable names

Four specific transformations were applied during this stage:

- (a) Reshaping (E&E Exports only): the original dataset was provided in wide format, with separate columns for each month within a year; this was reshaped ("melted") into a long format with a single Date column and a single export-value column, enabling it to be treated as a standard monthly time series.
- (b) Column selection (CPI Inflation): the original file contained inflation broken down by expenditure division (e.g. food, transport, housing); only the overall/composite CPI inflation column was retained for this study.
- (c) Renaming: all four datasets had their date and value columns renamed to a consistent naming convention (e.g. Date, E&E_Exports, CPI_Inflation) to simplify subsequent merging.
- (d) Date standardisation: all date fields, which were originally represented in differing formats across the four sources, were converted to a common datetime type at monthly resolution to allow consistent chronological ordering and merging.

4.4.1.2 Missing Value Treatment

Missing-value checks were performed after the individual dataset transformations.

Table 4:4 Missing Values in the Integrated Dataset

Variable	Observations	Missing Values
E&E Exports	423	0
CPI Inflation	423	0
Exchange Rate	423	0
U.S. Industrial Production Index	423	0

```
null_values_in_ee_exports = ee_exports_df.isnull().sum()
print("\nNull values in the transformed 'E&E Exports' DataFrame:")
print(null_values_in_ee_exports)
```

Figure 4:8 Missing Value Assessment of the E and E Exports

```
null_values_in_cpi_inflation = cpi_inflation_df.isnull().sum()
print("\nNull values in the transformed 'CPI Inflation' DataFrame:")
print(null_values_in_cpi_inflation)
```

Figure 4:9 Missing Value Assessment of the CPI Inflation

```
null_values_in_er_ = er_df.isnull().sum()
print("\nNull values in the transformed 'CPI Inflation' DataFrame:")
print(null_values_in_er_)
```

Figure 4:10 Missing Value Assessment of Exchange Rate

```

null_values_in_ipi_ = ipi_df.isnull().sum()
print("\nNull values in the transformed 'CPI Inflation' DataFrame:")
print(null_values_in_ipi_)

```

Figure 4:11 Missing Value Assessment of the Industrial Production Index

The results showed that the four prepared datasets contained no missing values in their respective analytical variables.

Therefore, no missing-value imputation was required at the dataset integration stage. This provided a complete monthly dataset for the subsequent exploratory analysis and feature engineering procedures.

4.4.1.3 Duplicate Assessment

A duplicate-value check was performed on the individual analytical variables.

```

duplicates_in_ee_exports = ee_exports_df.duplicated().sum()
print(f"\nNumber of duplicate rows in 'E&E Exports' DataFrame: {duplicates_in_ee_exports}")
if duplicates_in_ee_exports > 0:
    print("Duplicate rows found in 'E&E Exports' DataFrame:")
    display(ee_exports_df[ee_exports_df.duplicated()])
duplicates_in_date_column = ee_exports_df.duplicated(subset=['Date']).sum()
print(f"\nNumber of duplicate 'Date' entries in 'E&E Exports' DataFrame: {duplicates_in_date_column}")
if duplicates_in_date_column > 0:
    print("Rows with duplicate 'Date' entries in 'E&E Exports' DataFrame:")
    display(ee_exports_df[ee_exports_df.duplicated(subset=['Date'])])

```

Figure 4:12 Duplicate check for E and E exports

```

duplicates_in_cpi_inflation = cpi_inflation_df.duplicated().sum()
print(f"\nNumber of duplicate rows in 'CPI Inflation' DataFrame: {duplicates_in_cpi_inflation}")
if duplicates_in_cpi_inflation > 0:
    print("Duplicate rows found in 'CPI Inflation' DataFrame:")
    display(cpi_inflation_df[cpi_inflation_df.duplicated()])
duplicates_in_date_column_cpi = cpi_inflation_df.duplicated(subset=['Date']).sum()
print(f"\nNumber of duplicate 'Date' entries in 'CPI Inflation' DataFrame: {duplicates_in_date_column_cpi}")
if duplicates_in_date_column_cpi > 0:
    print("Rows with duplicate 'Date' entries in 'CPI Inflation' DataFrame:")
    display(cpi_inflation_df[cpi_inflation_df.duplicated(subset=['Date'])])

```

Figure 4:13 Duplicate check for CPI Inflation

```

duplicates_in_er_df = er_df.duplicated().sum()
print(f"\nNumber of duplicate rows in 'Exchange Rate' DataFrame: {duplicates_in_er_df}")
if duplicates_in_er_df > 0:
    print("Duplicate rows found in 'Exchange Rate' DataFrame:")
    display(er_df[er_df.duplicated()])
duplicates_in_date_column_er = er_df.duplicated(subset=['Date']).sum()
print(f"\nNumber of duplicate 'Date' entries in 'Exchange Rate' DataFrame: {duplicates_in_date_column_er}")
if duplicates_in_date_column_er > 0:
    print("Rows with duplicate 'Date' entries in 'Exchange Rate' DataFrame:")
    display(er_df[er_df.duplicated(subset=['Date'])])

```

Figure 4:14 Duplicate check for Exchange Rate

```

duplicates_in_ipi_df = ipi_df.duplicated().sum()
print(f"\nNumber of duplicate rows in 'US Industrial Production Index' DataFrame: {duplicates_in_ipi_df}")
if duplicates_in_ipi_df > 0:
    print("Duplicate rows found in 'US Industrial Production Index' DataFrame:")
    display(ipi_df[ipi_df.duplicated()])
duplicates_in_date_column_ipi = ipi_df.duplicated(subset=['Date']).sum()
print(f"\nNumber of duplicate 'Date' entries in 'US Industrial Production Index' DataFrame: {duplicates_in_date_column_ipi}")
if duplicates_in_date_column_ipi > 0:
    print("Rows with duplicate 'Date' entries in 'US Industrial Production Index' DataFrame:")
    display(ipi_df[ipi_df.duplicated(subset=['Date'])])

```

Figure 4:15 Duplicate check for U.S. Industrial Production Index

The results indicated that E&E Exports, CPI inflation, exchange rate and U.S. IPI contained no duplicated values.

4.4.2 Data Integration Results

After the individual datasets had been cleaned and standardised, they were integrated using the monthly date as the common reference key. A temporary Date_YM variable in YYYY-MM format was created for each dataset to ensure that observations from the same month were correctly aligned.

The E&E export dataset was used as the starting dataset and subsequently combined with CPI inflation, exchange rate and U.S. IPI data. After integration, the temporary Date_YM variable was removed, the date was converted into a datetime index, and the observations were sorted chronologically.

```

# Perform merges
# Start with E&E Exports
merged_df = ee_exports_df.copy()
# Merge with CPI Inflation
merged_df = pd.merge(merged_df, cpi_inflation_df[['Date_YM', 'CPI Inflation']], on='Date_YM', how='outer')
# Merge with Exchange Rate
merged_df = pd.merge(merged_df, er_df[['Date_YM', 'Exchange Rate']], on='Date_YM', how='outer')
# Merge with Industrial Production Index
merged_df = pd.merge(merged_df, ipi_df[['Date_YM', 'US Industrial Production Index']], on='Date_YM', how='outer')
merged_df['Date'] = pd.to_datetime(merged_df['Date_YM'])
merged_df = merged_df.drop(columns=['Date_YM'])
merged_df = merged_df.sort_values('Date').set_index('Date')
print("Merged DataFrame Head:")
# Create a temporary DataFrame for display with the index formatted as 'YYYY-MM'
display_df_head = merged_df.head().copy()
display_df_head.index = display_df_head.index.strftime('%Y-%m')
display(display_df_head)

```

Figure 4:16 Merging Dataset

Also, it was checked if the merged dataset contained any missing values and checked for duplicates to ensure if any inconsistencies was resulted but there was no missing values and duplicates found.

```

null_values_in_merged_=merged_df.isnull().sum()
print(null_values_in_merged_)

```

Figure 4:17 Null value checking in merged dataset

Missing-value assessment of the integrated dataset showing zero missing values across all analytical variables.

```

for column in merged_df.columns:
    if merged_df[column].duplicated().any():
        print(f" Column '{column}' has duplicate values.")
    else:
        print(f" Column '{column}' has no duplicate values.")

```

Figure 4:18 Duplicate checking in merged dataset

Duplicate observation assessment of the integrated dataset showing no duplicated rows or monthly date entries.

The resulting integrated dataset contained 423 monthly observations from January 1990 to March 2025 and four analytical variables.

Table 4:5 Integrated Dataset

Variable	Description	Observations
E&E Exports	Malaysia's monthly E&E export value	423
CPI Inflation	Monthly CPI inflation	423
Exchange Rate	MYR/USD exchange rate	423
U.S. Industrial Production Index	U.S. industrial production index	423

Each of the four datasets contained 423 monthly observations immediately prior to merging. After the successive left-join operations on the common Date_YM key, the resulting integrated dataset also contained 423 monthly observations — confirming that no observations were lost due to unmatched dates during integration, since all four sources shared an identical January 1990 – March 2025 coverage period at monthly frequency.

The integration results confirmed that all four variables were successfully aligned at a monthly frequency and that no missing values were introduced during the integration process. The resulting dataset therefore provided the basis for the exploratory data analysis and subsequent feature construction.

4.4.3 Original Integrated Dataset

The original integrated dataset consisted of 423 monthly observations covering the period from January 1990 to March 2025. The dataset was obtained by integrating the E&E export series with the three selected macroeconomic indicators using the monthly date as the common reference key.

The E&E export variable represents the target variable of the study, while inflation, the MYR/USD exchange rate and the U.S. Industrial Production Index

represent the initial explanatory variables. At this stage, the dataset represented the raw integrated data before the construction of lagged, rolling, growth-rate and time-based features required for the machine-learning models.

Table 4:6 Original Integrated Dataset

Item	Description
Target variable	E&E Exports
Macroeconomic variables	Inflation, MYR/USD Exchange Rate, U.S. Industrial Production Index
Frequency	Monthly
Start date	January 1990
End date	March 2025
Number of observations	423
Initial dataset type	Multivariate time-series dataset

The 35-year observation period provided a broad historical representation of Malaysia's E&E export performance across different economic conditions. It included periods of economic expansion and contraction, the 2008 Global Financial Crisis, the COVID-19 pandemic, and the subsequent economic recovery. These periods provided variation in the E&E export series for subsequent exploratory analysis and forecasting model development.

4.4.4 Dataset Structure and Variables

The original integrated dataset was structured as a multivariate monthly time-series dataset, where each row represented one monthly observation and each column represented the target variable or an explanatory variable. The Date variable served as the common temporal index for integrating the datasets and maintaining the chronological order of observations.

The integrated dataset has dimensions of 423 rows $\times$ 5 columns (one Date column plus four analytical variables), covering January 1990 to March 2025. A sample of the first and last three rows is shown below:

Table 4:7 First and Last 3 Row of the Dataset

Date	E&E Exports	CPI Inflation	Exchange Rate	US Industrial Production Index
01-01-1990	1010.235516	0.7	2.7041	61.729
01-02-1990	1034.73632	0.5	2.7032	62.2896
01-03-1990	1249.33347	-0.2	2.717	62.5999
01-01-2025	40494.05477	0.1	4.462	100.0647
01-02-2025	36504.76002	0.4	4.4408	101.0993
01-03-2025	45813.71207	0	4.4325	101.0404

Because all four source variables were merged on a common Date_YM (year-month) key rather than an exact calendar date, every row of the integrated dataset is confirmed to represent the same calendar month across all four variables — i.e., the E&E export, inflation, exchange-rate and IPI values in any given row all correspond to the identical month.

Table 4:8 Structure of Original Dataset

Variable	Description	Unit	Data Type	Role	Source
Date	Monthly observation date	YYYY-MM	Date/Time	Time index	—
E&E Exports	Malaysia's monthly E&E export value	RM million	Numerical	Target	Ministry of Economy Malaysia
CPI Inflation	Monthly inflation indicator	% change	Numerical	Descriptive Analysis	DOSM
MYR/USD Exchange Rate	Exchange rate between MYR and USD	MYR per USD	Numerical	Descriptive Analysis	FRED
U.S. Industrial Production Index	Indicator of U.S. industrial activity	Index (2017=100)	Numerical	Descriptive Analysis	FRED

The target variable was Malaysia's E&E export value, while the initial variables consisted of CPI inflation, the MYR/USD exchange rate and the U.S. Industrial Production Index were used for descriptive analysis. These variables provided information on domestic price conditions, exchange-rate movements and external industrial activity, respectively.

4.5 Exploratory Data Analysis

This section presents the exploratory data analysis conducted on Malaysia's E&E export series and the selected macroeconomic predictors, following the data

preprocessing and integration described in Section 4.4. The analysis begins with an examination of the historical trend, variability, and unusual observations in the E&E export series (Section 4.5.1), followed by seasonal decomposition to separate trend, seasonal, and residual components (Section 4.5.2). Autocorrelation analysis is then conducted to assess the temporal dependence structure of the series and inform subsequent ARIMA model specification (Section 4.5.3), before examining seasonality and recurring patterns specific to E&E exports (Section 4.5.4). Descriptive statistics summarising the distribution of all variables are presented in Section 4.5.5, and correlation analysis between E&E exports and the macroeconomic predictors is conducted in Section 4.5.6 to identify relationships that may support the forecasting analysis in later chapters.

4.5.1 Historical Trend, Variations and Unusual Observations

To examine the long-run behaviour of Malaysia's E&E exports, the monthly export series was plotted over the full January 1990 – March 2025 study period. This visualisation provides the basis for identifying the overall trend, structural shifts and periods of heightened volatility discussed below.

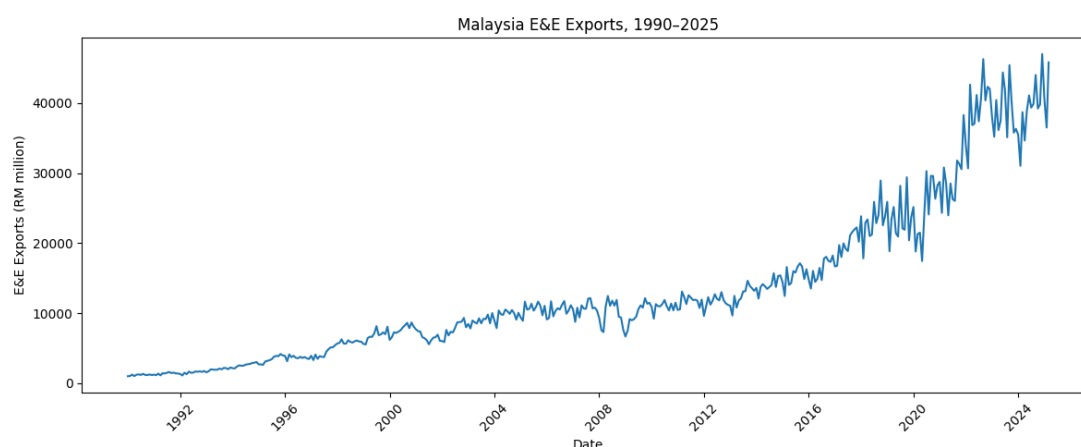


Figure 4:19 Monthly E&E Export Values, January 1990 – March 2025

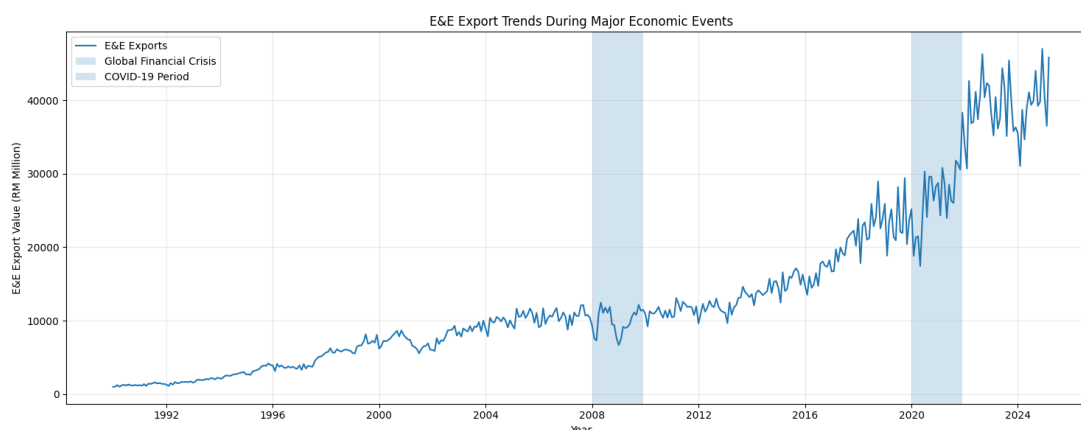


Figure 4:20 E&E Export Trends During Major Economic Events

Malaysia's monthly E&E exports grew from RM1,010.24 million in January 1990 to RM45,813.71 million in March 2025, an overall increase of 4,434.95% across the 35-year study period (Figure 4:22). This growth was not linear: the series shows a gradual, relatively low-volatility rise through the 1990s and 2000s, followed by markedly higher month-to-month volatility from around 2016 onward, with values oscillating between roughly RM30,000 million and RM47,000 million rather than following a smooth upward path.

Two periods of unusual behaviour stand out. During the 2008–09 Global Financial Crisis, mean monthly exports fell to RM9,942.18 million (std. dev. RM1,701.98 million) well below the RM12,530.48 million average of the surrounding normal period with year-on-year growth averaging -2.41% during this period, though individual months still ranged as widely as -32.31% to $+49.75\%$, reflecting sharp swings as exports collapsed and partially recovered within the crisis window. During the COVID-19 period (2020–21), exports instead rose sharply to a mean of RM26,910.16 million, with year-on-year growth averaging $+12.77\%$ comparable to the $+13.14\%$ average growth of normal periods indicating that E&E exports were resilient to, and arguably benefited from, pandemic-era global electronics demand rather than experiencing the contraction typically associated with the pandemic in other sectors. These two episodes represent genuine structural shifts in the series rather than data anomalies, and are retained in the dataset because they carry real information about how the series behaves under stress information that a well-specified ARIMA

model must be able to accommodate, and that later motivates checking model residuals for remaining structure around these periods (Section 4.5.3).

4.5.2 Seasonal Decomposition

To formally assess whether Malaysia's E&E export series contains a systematic seasonal component in addition to its trend, the series was decomposed using classical additive decomposition with a 12-month period (Figure 4:21):

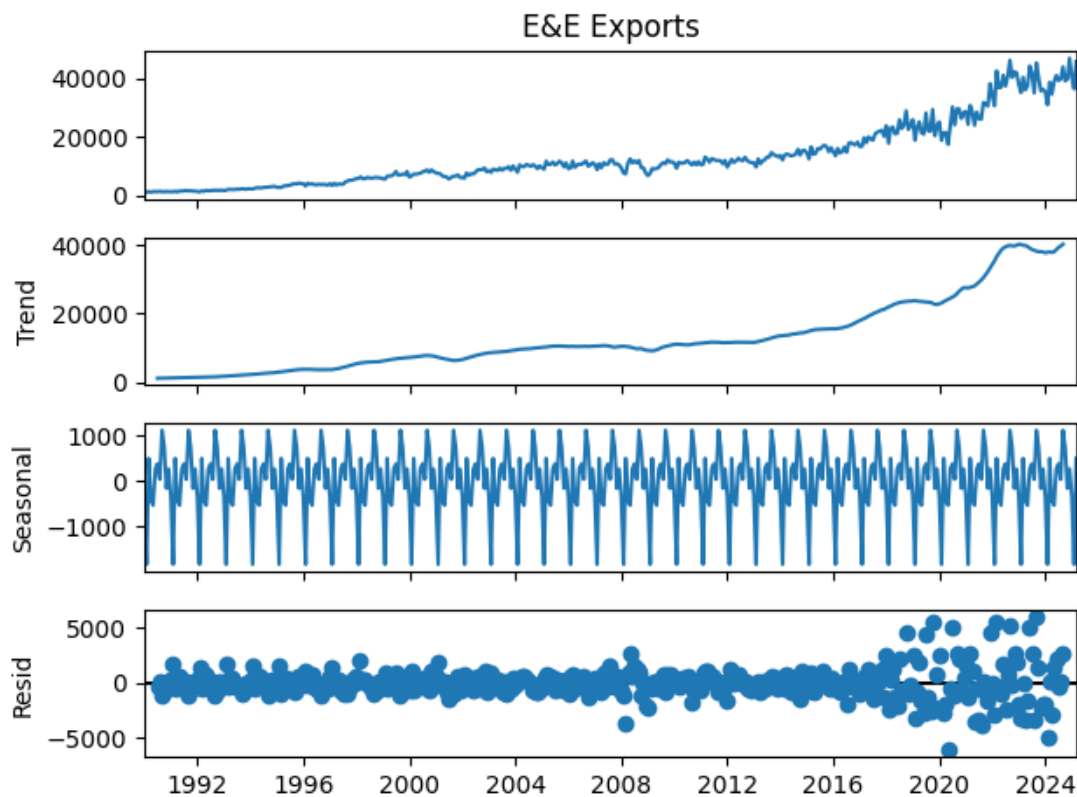


Figure 4:21 Seasonal Decomposition of E and E Exports

The decomposition separates the series into three components. The trend component closely tracks the raw series' long-run trajectory identified in Section 4.5.1 a gradual rise through the 1990s and 2000s that steepens noticeably from around 2016 onward, confirming that most of the series' movement is attributable to sustained long-run growth rather than short-term fluctuation.

The seasonal component is small and highly regular, oscillating within a consistent band of approximately $\pm 1,000$ RM million around zero throughout the entire 35-year period — a stable, repeating annual pattern consistent with the monthly averages reported in Table 4:X (troughs around February, peaks around March and September). Because this seasonal amplitude ($\pm 1,000$) is small relative to the series' overall scale (which grew from $\sim 1,000$ to $\sim 45,000$ RM million), the seasonal effect is real but modest in absolute terms compared to the dominant trend.

The residual component remains tightly clustered around zero for most of the series but visibly widens from around 2016–2018 onward, with noticeably larger and more scattered residuals through 2020–2024 — indicating that the model's fixed 12-month additive seasonal pattern explains the earlier, calmer portion of the series well, but leaves more unexplained variation in the recent, more volatile period. This residual widening is consistent with the increased month-to-month volatility already noted in Section 4.5.1, and signals that any ARIMA specification for this series should be checked for adequacy separately across both the earlier low-volatility period and the recent high-volatility period during diagnostic checking

4.5.3 Autocorrelation Analysis

To further investigate the series' autocorrelation structure — both to confirm the seasonality finding above and to inform ARIMA order selection in Chapter 3 — the autocorrelation function (ACF) and partial autocorrelation function (PACF) of the E&E export series were plotted up to 24 lags (Figure 4:22) and (Figure 4:23):

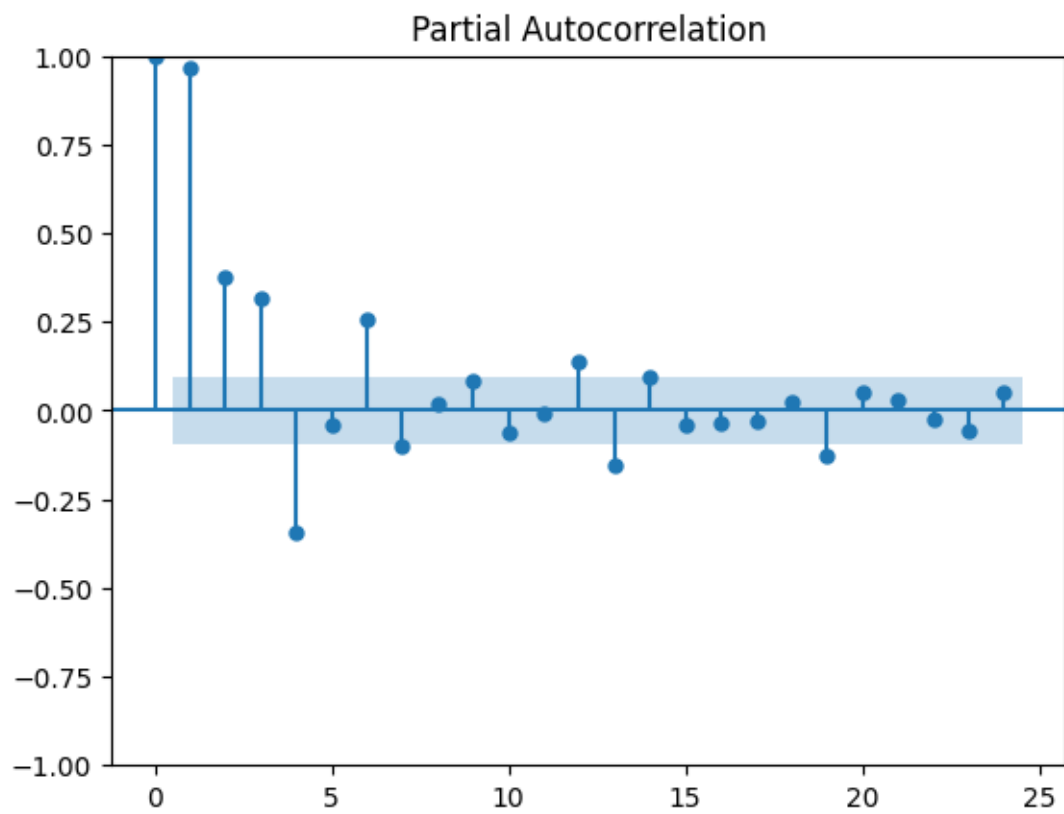


Figure 4:22 Partial Autocorrelation of E and E export

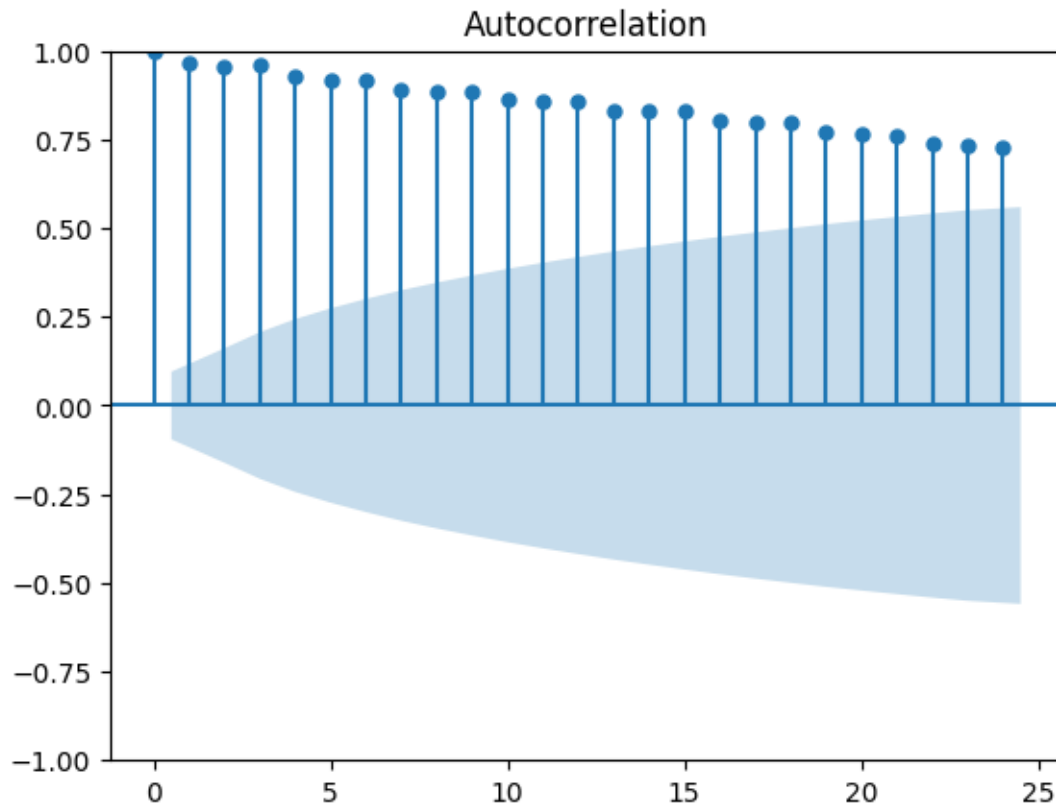


Figure 4:23 Autocorrelation of E and E export

The ACF plot shows very high autocorrelation at lag 1 (approximately 0.97) that decays slowly and smoothly across all 24 lags, remaining as high as approximately 0.73 even at lag 24. This slow, gradual decay — rather than a sharp cutoff after a small number of lags — is a classic signature of a non-stationary, trend-dominated series (Box and Jenkins, 1970), and provides direct empirical confirmation that E&E exports require differencing before ARIMA modelling can proceed (Section 3.X). Notably, the ACF does not show a distinct secondary spike at lag 12 or lag 24 — if a strong seasonal effect were present, autocorrelation would be expected to rise again at those seasonal lags rather than decaying smoothly. This is consistent with the decomposition finding above: seasonality exists but is modest relative to the strong trend, and the trend's autocorrelation dominates the ACF to the point of masking any seasonal signal at this level of analysis.

The PACF plot shows a markedly different pattern: a large partial autocorrelation at lag 1 (approximately 0.97), dropping sharply to approximately 0.38

at lag 2, and continuing to decline with a notable negative dip to approximately -0.35 at lag 4, before settling within the approximate 95% confidence bounds from around lag 7 onward. This sharp cutoff after the first one to two lags — in contrast to the ACF's slow decay — is consistent with an autoregressive process of low order (AR(1) or AR(2)) once the series is appropriately differenced, and provides useful preliminary evidence for the candidate AR order to be tested during formal order selection in Chapter 3 (alongside AIC-based comparison of candidate specifications). The negative partial autocorrelation at lag 4 is comparatively minor and falls within a plausible range for sampling noise, but is noted here for completeness and will be checked against residual diagnostics once a candidate ARIMA model is fitted.

4.5.4 E&E Export Seasonality and Patterns

With the date index fixed, the monthly average line chart now shows a genuine seasonal pattern. Exports dip to their lowest point in February (~RM11,500 million) and are consistently higher in March and September (~RM14,000–14,300 million), with a secondary lift in October and December. The shape is a jagged zig-zag rather than a smooth curve no single long stretch of the year is uniformly high or low, but Feb/Apr/May stand out as the weaker months and Mar/Sep/Oct/Dec as the stronger ones.

To assess whether E&E exports exhibit recurring seasonal patterns, average export values were calculated for each calendar month across the full study period and plotted below.

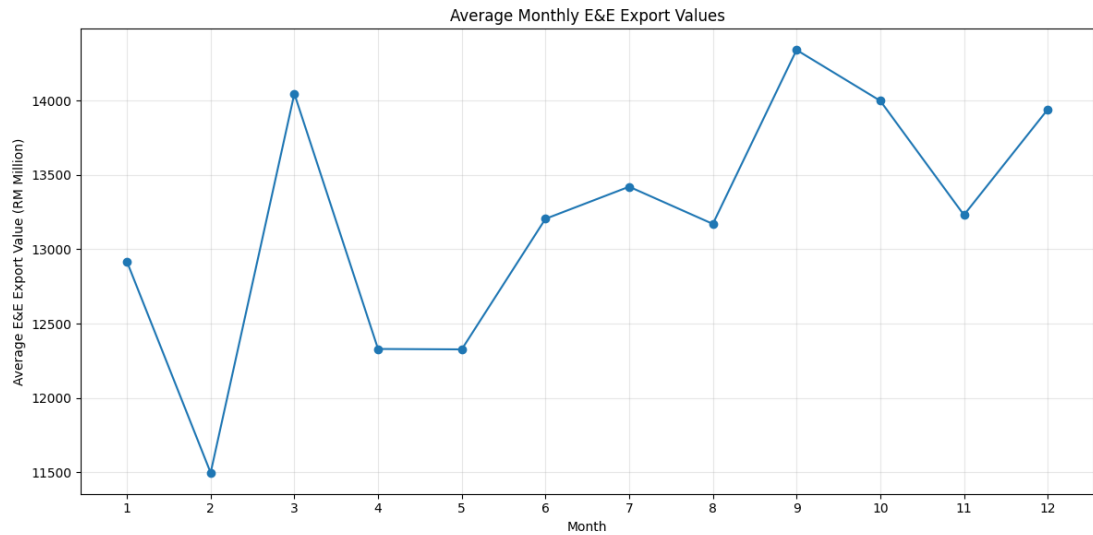


Figure 4:24 Average Monthly E&E Export Values

The monthly boxplot tells a complementary story: every month's box sits at a similar height (medians cluster around RM9,000–11,500 million) and every month shows a long upper whisker with scattered high outliers stretching to RM40,000+ million. This means the seasonal effect is modest relative to the *trend* effect — the difference between a strong month and a weak month is small compared to how much a given month's exports have grown across the 35-year span. February visibly has the lowest median and a noticeably shorter box than the rest, confirming it as the weakest month overall.

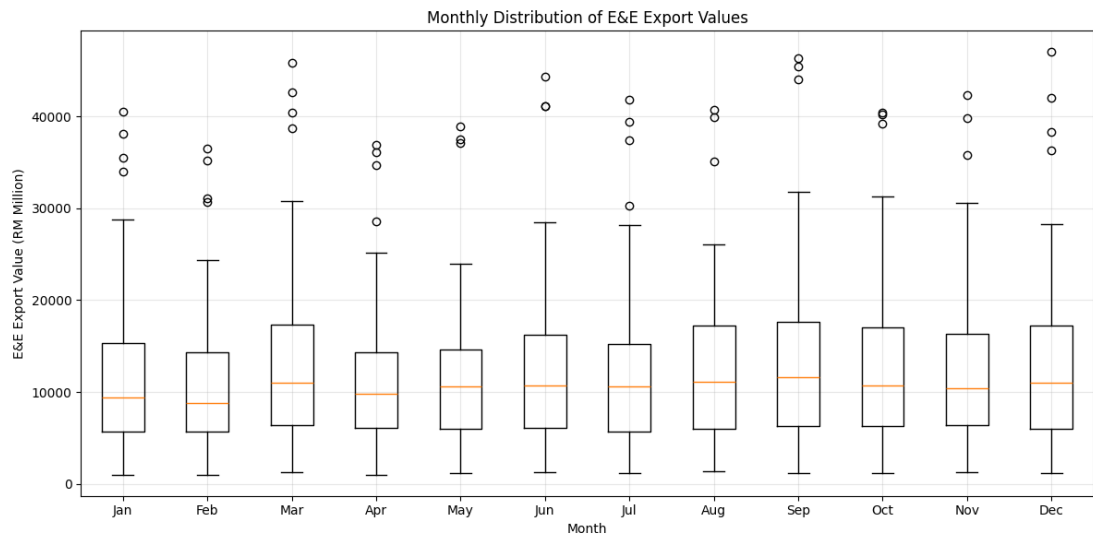


Figure 4:25 Monthly Distribution of E&E Export Values

The monthly coefficient of variation ranges narrowly from 0.785 (August) to 0.864 (March) across all twelve months (Table 4:X), indicating broadly similar relative dispersion across the calendar year rather than any month standing out as substantially more or less variable than the others. As with the overall series-level CV of 0.825 discussed in Section 4.5.5, these monthly CV values describe dispersion relative to each month's own mean across the 35-year sample — they do not describe month-to-month change. The appropriate measure of typical monthly volatility is the mean absolute month-to-month percentage change, computed directly on the full 423-observation series:

```
monthly_pct_change = df['E&E Exports'].pct_change().abs() * 100
mean_abs_pct_change = monthly_pct_change.mean()
print(mean_abs_pct_change)
```

Figure 4:26 Code Snippet for showing the monthly E&E Export change

Malaysia's E&E exports change by an average of 9.21% from one month to the next (in either direction) a figure that reflects genuine, moderate month-to-month volatility, and is the correct number to cite when discussing typical monthly fluctuation, rather than the CV.

4.5.5 Descriptive Statistics

This section presents descriptive statistics for Malaysia's E&E export series and the three macroeconomic predictor variables, summarising their central tendency, dispersion, and distributional shape across the full study period. These statistics complement the trend, seasonality, and autocorrelation analyses presented in the preceding sections by providing a concise numerical characterisation of each variable prior to model development. Table 4:9 reports the mean, median, standard deviation, minimum, maximum, skewness, and coefficient of variation (CV) for each variable, with the results interpreted below.

Table 4:9 Descriptive statistics table and interpretation

Variable	Mean	Median	Std. Dev.	Min	Max	Skewness	CV
E&E Exports (RM million)	13,199.49	10,528.50	10,892.02	1,010.24	47,002.31	1.31	0.825
CPI Inflation (%)	0.21	0.20	0.48	-2.70	4.00	1.92	—
Exchange Rate	3.54	3.76	0.64	2.44	4.77	-0.20	0.18
U.S. IPI	90.09	94.08	12.92	60.33	104.10	-1.08	0.14

The coefficient of variation is not reported for CPI Inflation. CV is calculated as the standard deviation divided by the mean, and is only meaningful for variables where the mean is a meaningful, non-zero reference point. CPI Inflation has a mean of just 0.21% a value very close to zero so dividing its standard deviation (0.48) by this near-zero mean produces a CV of approximately 2.3, an artificially inflated figure that reflects the smallness of the denominator rather than genuine instability in the series. Because CPI Inflation also takes both positive and negative values (ranging from -2.70% to 4.00%), a ratio-based measure like CV is not statistically appropriate for it in the first place. For this reason, only the exchange rate, U.S. IPI and E&E exports —

all strictly positive, ratio-scale variables are reported with a CV, while CPI Inflation is better characterised by its standard deviation and range alone.

The mean E&E export value of RM13,199.49 million, well above the median of RM10,528.50 million, together with a skewness of 1.31, confirms that the distribution is right-skewed — consistent with a series that spent most of its 35-year history at lower export levels before growing substantially in its later years. The standard deviation of RM10,892.02 million relative to the mean produces a coefficient of variation (CV) of 0.825. This CV describes the dispersion of monthly export values relative to their mean across the entire 35-year series — it does not mean that exports change by 82.5% from one month to the next. A CV of this magnitude for a strongly trending series simply reflects the fact that export levels near the start of the series (RM1,010.24 million) are vastly different from levels near the end (RM45,813.71 million) an overall increase of 4,434.95% rather than being a measure of month-to-month change. The more directly relevant measure of monthly variability is the mean absolute month-to-month percentage change in exports, computed on the full 423-observation series (Section 4.5.4), which was 9.21% the figure that should be used when discussing typical monthly volatility, rather than the CV.

To assess the distributional characteristics of each variable including skewness and the presence of extreme values histograms were plotted for E&E exports and each of the three macroeconomic indicators.

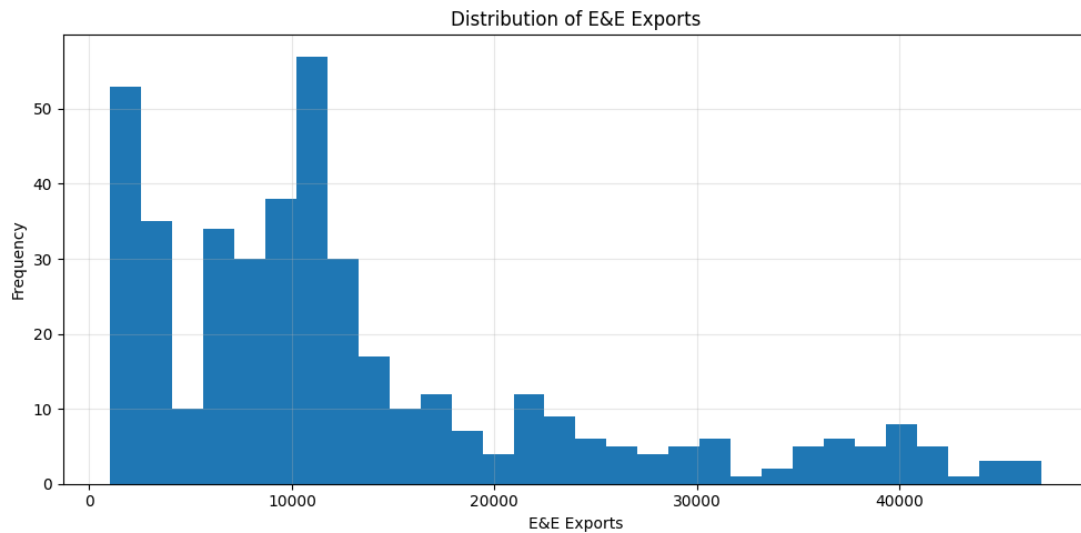


Figure 4:27 Distribution of E&E Exports

E&E Exports histogram: strongly right-skewed — a tall cluster of bars sits in the RM0–15,000 million range, with a long thinning tail stretching to RM47,000 million. This visually matches the skewness value of 1.31 and reflects the trend structure: many years of lower values, fewer years of very high values.

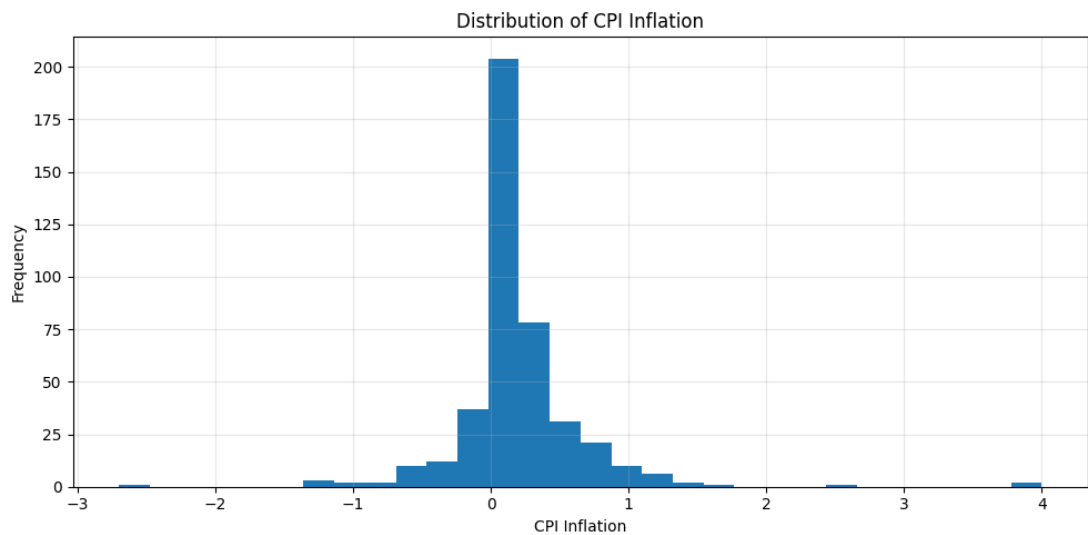


Figure 4:28 Distribution of CPI Inflation

CPI Inflation histogram: tightly clustered near 0–1%, with a handful of extreme outlier bars pulling the shape rightward these correspond to the inflation spikes visible around 2002, 2008, and 2022 in the earlier line chart.

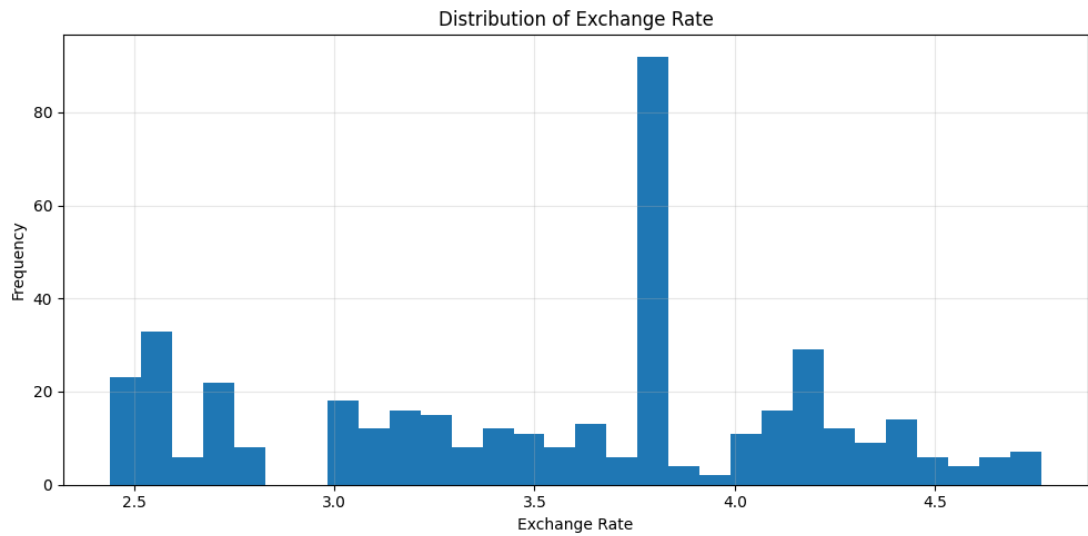


Figure 4:29 Distribution of Exchange Rate

Exchange Rate histogram: loosely bimodal one cluster around 2.5–3.0 (the pre-1998 peg era) and another around 3.5–4.7 (the post-float era), rather than one smooth bell curve.

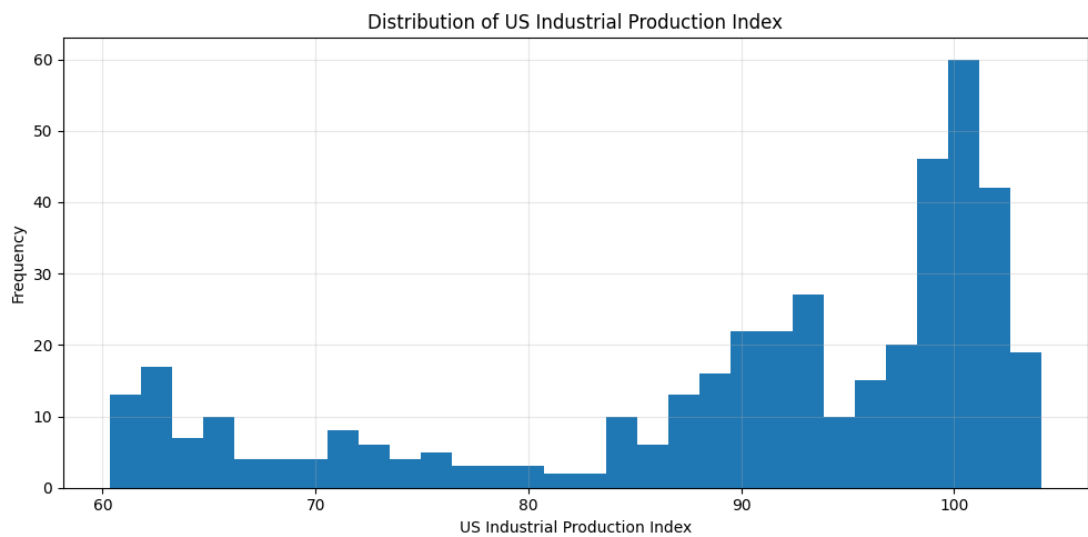


Figure 4:30 Distribution of U.S. Industrial Production Index

US Industrial Production Index histogram: left-skewed, with most values bunched at the higher end (85–104) and a thinner tail of lower values, reflecting the index spending most of its time near historical highs and dipping only briefly during 2008 and 2020.

E&E Exports (skewness = 1.31) and CPI Inflation are both right-skewed, while U.S. IPI is left-skewed.

4.5.6 Correlation Analysis

To examine the linear relationships between E&E exports and the selected macroeconomic indicators, a correlation matrix was constructed and visualised as a heatmap.

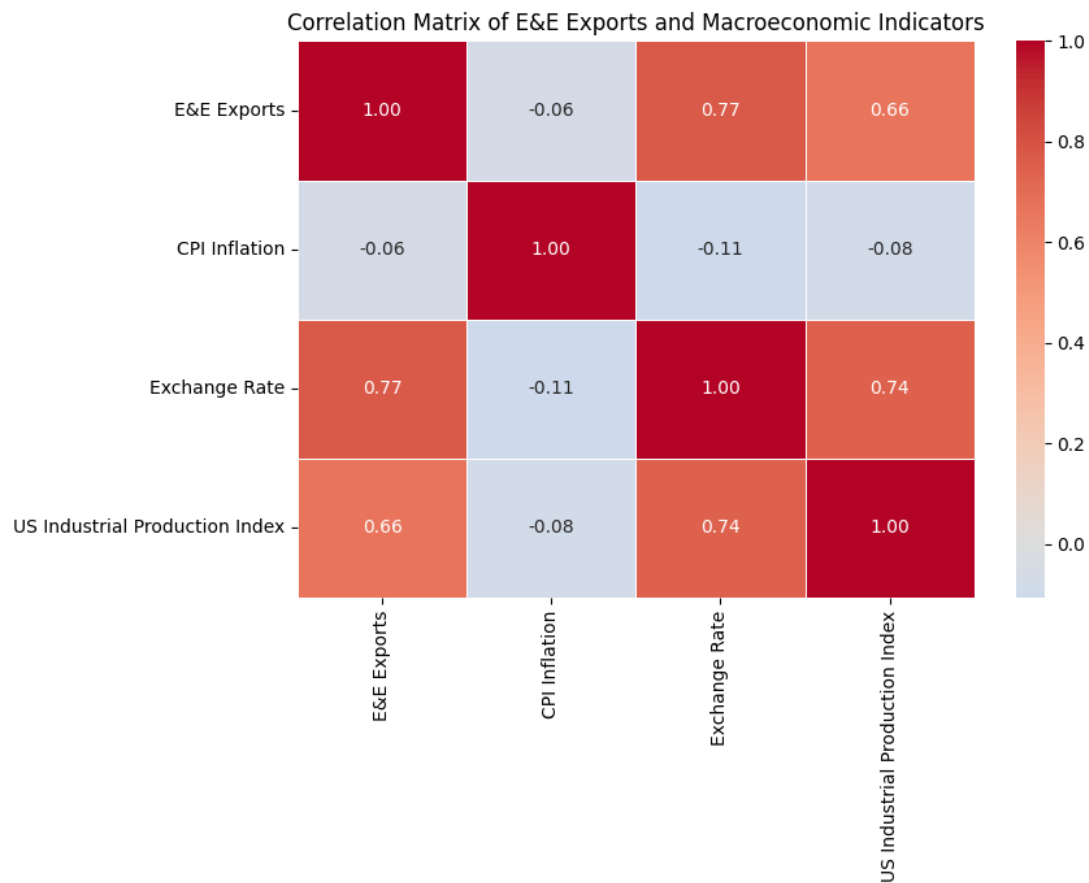


Figure 4:31 Correlation Matrix of E&E Exports and Macroeconomic Indicators

The heatmap makes the relationships easy to scan: the darkest/warmest cells are E&E Exports–Exchange Rate (0.77) and Exchange Rate–U.S. Industrial Production (0.74), both showing strong positive correlations. E&E Exports–U.S. Industrial Production (0.66) is a slightly lighter red, indicating a moderately strong positive correlation. By contrast, the entire CPI Inflation row/column is pale, cool-toned near zero, visually standing out as having relatively weak linear relationships with the other variables, while the exchange rate, U.S. Industrial Production and E&E exports show stronger positive associations.

4.6 Chapter Summary

This chapter presented the results of the data preparation and exploratory data analysis conducted for the Malaysia E&E export forecasting study. The original integrated dataset consisted of 423 monthly observations from January 1990 to March 2025, containing E&E export values, CPI inflation, MYR/USD exchange rate, and the U.S. Industrial Production Index, successfully cleaned, standardised, and integrated using a common monthly date index.

The exploratory analysis examined the historical trend and structural shifts in the E&E export series, including its behaviour during the 2008–09 Global Financial Crisis and the COVID-19 period. Seasonal decomposition confirmed that the series' movement is dominated by a strong long-run trend, with a modest but genuine annual seasonal component; autocorrelation analysis (ACF and PACF) further showed a slowly decaying ACF consistent with non-stationarity, and a PACF cutoff after the first one to two lags consistent with a low-order autoregressive process — both findings that directly inform the ARIMA model identification process presented in Chapter 3. Descriptive statistics and distributional analysis characterised each variable's central tendency, dispersion, and skewness, and clarified the correct interpretation of the coefficient of variation for a strongly trending series. Finally, correlation analysis addressed RQ1 by examining the descriptive relationship between E&E exports and the three macroeconomic indicators, finding the strongest association with the exchange rate ($r = 0.767$) and U.S. Industrial Production Index ($r = 0.664$), and a negligible relationship with CPI inflation ($r = -0.063$); these relationships are reported for contextual understanding only and are not used as inputs to the ARIMA forecasting model, which is developed purely from the export series' own historical structure.

Together, these findings particularly the evidence of non-stationarity and low-order autocorrelation establish the empirical basis for the ARIMA model identification and development process presented in Chapter 3.

CHAPTER 5

IMPLEMENTATION

5.1 Introduction

This chapter presents the technical implementation of the ARIMA forecasting model developed for this study, following the model-identification steps outlined in Chapter 3: stationarity testing, order selection, model fitting, diagnostic checking, and forecast generation. As supplementary work, Random Forest, SVR, and XGBoost were also implemented and evaluated against the ARIMA model; their implementation, hyperparameter tuning, and detailed results are presented in Appendix A rather than in this chapter, consistent with the study's primary focus on ARIMA-based forecasting.

5.2 Forecasting System Architecture

The forecasting system architecture (Figure 5:1) was simplified to reflect the ARIMA-focused pipeline.

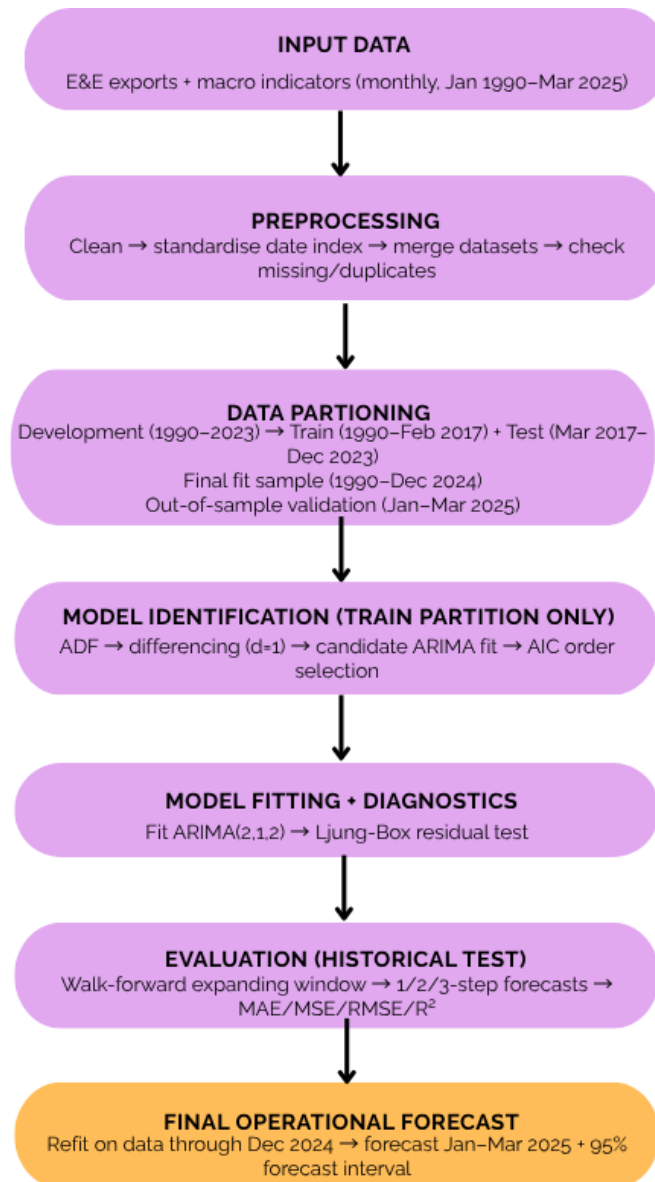


Figure 5:1 Forecasting System Architecture

5.3 Stationarity Testing

Before ARIMA can be applied, the E&E export series must be tested for stationarity. Chapter 4's ACF analysis (Section 4.5.3) already showed slow, gradual autocorrelation decay informal visual evidence of non-stationarity which was confirmed formally using the Augmented Dickey-Fuller (ADF) test.

```
adf_raw = adfuller(train)
print(f"ADF (raw): stat={adf_raw[0]:.4f} p={adf_raw[1]:.4f}")
```

Figure 5:2 ADF Test applying first order differencing

After applying first-order differencing, the ADF test was repeated on the differenced series:

```
diff = train.diff().dropna()
adf_diff = adfuller(diff)
print(f"ADF (diff): stat={adf_diff[0]:.4f} p={adf_diff[1]:.4f}")
```

Figure 5:3 ADF Test applying differenced

The ADF test was applied to the raw E&E export series in the initial training partition (326 observations; January 1990–February 2017). The test statistic was 0.3140 with a p-value of 0.9780, meaning the null hypothesis of a unit root cannot be rejected. This confirms that the raw series is non-stationary. After applying first-order differencing, the ADF statistic dropped to -6.7915 with a p-value below 0.0001, strongly rejecting the unit-root null and confirming that the differenced series is stationary. Therefore, $d = 1$ was used for ARIMA model specification.

5.4 ARIMA Order Selection

Following the Box-Jenkins identification approach, six candidate ARIMA(p,1,q) specifications were fitted to the training series and compared using the Akaike Information Criterion (AIC).

Table 5:1 AIC Order Selection

Order	AIC	BIC
(2,1,2)	5289.81	5308.73
(2,1,1)	5293.26	5308.39
(3,1,1)	5293.50	5312.42
(2,1,0)	5293.90	5305.25
(1,1,1)	5302.30	5313.65
(1,1,0)	5313.63	5321.20

ARIMA(2,1,2) was selected as the final model specification, achieving the lowest AIC among all candidates tested. This choice is consistent with the PACF evidence from Section 4.5.3, which suggested a low-order autoregressive process (AR(1) or AR(2)) after differencing, and adds a second-order moving-average component that further reduces AIC relative to the simpler (2,1,0) and (1,1,1) alternatives.

When the same six candidate specifications were re-fitted on the full available series through December 2024 (420 observations, rather than the original 326-observation training set), ARIMA(2,1,2) again achieved the lowest AIC (7,302.96), confirming that this order selection is robust to the additional data accumulated between the initial training period and the most recent observations.

Because AIC depends on both model fit and sample size, the development-sample AIC values (326 observations) and final-sample AIC values (420 observations) are reported separately and are not directly comparable to each other.

5.5 Diagnostic Checking

Once ARIMA(2,1,2) was fitted, its residuals were checked to confirm the model adequately captures the series' structure, leaving only white noise.

```
lb = acorr_ljungbox(final_model.resid, lags=[10,20], return_df=True)
print(lb)
```

Figure 5:4 Diagnostic Checking

Residual diagnostics were conducted using the Ljung–Box test to assess whether the fitted model residuals approximate white noise. For the development-stage ARIMA(2,1,2) model fitted on the initial 326-observation training partition, the Ljung–Box test returned statistically significant results at both lag 10 ($p = 2.89 \times 10^{-5}$) and lag 20 ($p = 1.03 \times 10^{-12}$), indicating remaining autocorrelation in the residuals. Repeating the residual diagnostic after fitting ARIMA(2,1,2) on the full operational training sample through December 2024 (420 observations) again yielded significant Ljung–Box statistics at lag 10 ($p = 2.82 \times 10^{-8}$) and lag 20 ($p = 6.87 \times 10^{-11}$). This indicates that while ARIMA(2,1,2) is the best-performing model among the tested non-seasonal candidates, some residual time-series structure remains unmodelled. This limitation is documented and supports the recommendation to evaluate a seasonal ARIMA (SARIMA) extension in future work.

5.6 Model Fitting and Walk-Forward Forecasting Procedure

Rather than fitting ARIMA(2,1,2) once and forecasting the entire test period from a single origin, this study used a walk-forward (expanding-window) forecasting procedure: at each point in the 82-observation test period, the model is refit on all data available up to that point, then used to generate a 3-step-ahead forecast, before the next actual observation is added to the history and the process repeats.

```

orders = [(1,1,0),(1,1,1),(2,1,0),(2,1,1),(2,1,2),(3,1,1)]
results = []
for o in orders:
    m = ARIMA(train, order=o).fit()
    results.append({'Order': o, 'AIC': m.aic})
res_df = pd.DataFrame(results).sort_values('AIC').reset_index(drop=True)
print(res_df)
best_order = res_df.iloc[0]['Order']

```

Figure 5:5 Selecting best order for model fit

```

final_model = ARIMA(train, order=best_order).fit()
print("AIC:", final_model.aic, " BIC:", final_model.bic)

```

Figure 5:6 Fitting ARIMA Model

The walk-forward process generated 82 one-month-ahead forecasts, 81 two-month-ahead forecasts, and 80 three-month-ahead forecasts across the historical test partition (March 2017–December 2023).

```

for horizon in range(1, 4):
    target_position = origin_number + horizon - 1

    if target_position < len(test_dev):
        target_date = test_dev.index[target_position]
        actual_value = test_dev.iloc[target_position]
        forecast_value = forecasts.iloc[horizon - 1]

        walk_forward_records.append({
            "Origin Date": history.index[-1],
            "Target Date": target_date,
            "Horizon": horizon,
            "Actual": actual_value,
            "Forecast": forecast_value,
            "Error": actual_value - forecast_value
        })

# Reveal only the next actual observation
next_actual = test_dev.iloc[[origin_number]]
history = pd.concat([history, next_actual])

```

Figure 5:7 Expanding-window walk-forward forecasting procedure used across the historical test partition

This approach is more realistic than a single static forecast because it mirrors how the model would actually be used in production: each month, as new actual export data becomes available, the model is refit and a fresh three-month forecast is generated, rather than relying on an increasingly stale single forecast made at the start of the test period.

5.7 Model Evaluation

Applying the walk-forward procedure across the 82-observation test period, ARIMA(2,1,2) achieved the following performance.

Table 5:2 Arima Model Evaluation

Horizon	MAE	MSE	RMSE	R ²
1-month	2,164.75	7,842,517.00	2,800.45	0.8738
2-month	2,347.59	8,828,279.00	2,971.24	0.8575
3-month	2,585.11	10,622,640.00	3,259.24	0.8266

Forecast accuracy declines as the horizon lengthens, with RMSE rising from RM2,800.45 million at one month to RM3,259.24 million at three months, and R² decreasing from 0.8738 to 0.8266.

5.8 Out-of-Sample Validation: January–March 2025

To assess how the final ARIMA (2,1,2) model performs on genuinely unseen data, the model was fitted on all observations through December 2024 (420 observations) and used to forecast three steps ahead — January, February, and March 2025 with these forecasts then compared against the actual values already available for this period.

```

forecast = final_model.get_forecast(steps=3)
mean_fc = forecast.predicted_mean
ci = forecast.conf_int(alpha=0.05)

```

Figure 5:8 Forecasting January -March 2025

Table 5:3 Out-of-sample forecast comparison, January–March 2025

Month	Actual (RM mn)	Forecast (RM mn)	95% CI	Error	Abs. % Error
Jan 2025	40,494.05	41,541.21	[38,663.02, 44,419.39]	−1,047.15	2.59%
Feb 2025	36,504.76	42,881.24	[39,582.49, 46,180.00]	−6,376.48	17.47%
Mar 2025	45,813.71	46,608.44	[42,850.88, 50,366.00]	−794.72	1.73%

Across these three months, the model achieved a mean absolute error of RM2,739.45 million and an RMSE of RM3,758.89 million. January and March forecasts were close to the actual values, with absolute percentage errors of 2.59% and 1.73% respectively, and both actuals fell comfortably within the model's 95% forecast interval. February, however, shows a substantially larger error of 17.47%, with the model overestimating exports by RM6,376.48 million; although the actual value fell just outside the lower bound of the 95% interval, the direction and rough scale of the miss are informative.

This pattern is consistent with the residual diagnostic finding in Section 5.5. December 2024 recorded an unusually high export value (RM47,002.31 million), and the model lacking any seasonal term to anticipate February's characteristic seasonal trough (the lowest-averaging month in the historical series, Section 4.5.4) — extrapolated continued strength from December's elevated level rather than anticipating the seasonal pullback that actually occurred. This is precisely the kind of error a non-seasonal ARIMA model is structurally prone to, and reinforces both the

residual diagnostic finding and the case for testing a SARIMA specification as future work.

January and March 2025 actual values fell within the 95% forecast interval, while February fell below the lower bound, indicating an unusually large negative shock relative to the model's expected range.

5.9 Power BI Dashboard Implementation

A consolidated one-page Power BI dashboard was developed to communicate the results of the E&E export forecasting system. The dashboard integrates the historical export series (January 1990–March 2025), the final ARIMA(2,1,2) forecast outputs and forecast intervals, and the model's performance results derived from walk-forward evaluation.

The dashboard presents five main components. First, KPI cards summarise the selected model specification (ARIMA(2,1,2)), the latest actual E&E export value, the latest forecast value, and the January–March 2025 validation error metrics (MAE and RMSE). Second, a time-series line chart shows the historical E&E export trend together with the January–March 2025 forecast and the 95% forecast interval to visualise forecast uncertainty. Third, an out-of-sample validation table reports the January–March 2025 actual values, forecasts, forecast intervals and absolute percentage errors, allowing users to identify months where the forecast deviates substantially from actual outcomes. Fourth, walk-forward evaluation visuals summarise the model's test-period performance across one-, two- and three-month horizons using RMSE and R^2 . Finally, the dashboard includes explanatory notes clarifying that macroeconomic indicators are provided for descriptive context only and are not ARIMA model inputs.

This dashboard therefore functions as an accessible interface for stakeholders to review the export trend, short-term forecast outlook, forecast uncertainty and model accuracy in a single view. The ARIMA modelling and evaluation were performed in

Python, and the resulting outputs were exported to CSV and imported into Power BI for visualisation.

5.10 Supplementary Machine-Learning Comparison

As supplementary work, Random Forest, SVR, and XGBoost were also implemented and evaluated against the ARIMA model developed above, using the same 80:20 chronological train-test split and evaluation metrics. The full implementation, hyperparameter tuning, and results of this comparison are presented in Appendix A, and are discussed briefly, for context, in Chapter 6.

5.11 Chapter Summary

This chapter details the ARIMA forecasting methodology applied to Malaysia's monthly E&E exports, using a dataset of 423 observations from January 1990 to March 2025. The data was divided into a training sample of 326 observations and a test sample of 82. Initial stationarity testing indicated the series was non-stationary ($ADF = 0.3140$, $p = 0.9780$) but became stationary after first-order differencing ($ADF = -6.7915$, $p < 0.0001$). Six ARIMA ($p,1,q$) models were evaluated, with ARIMA (2,1,2) selected for its lowest AIC value (5289.81). Residual checks revealed significant autocorrelation, suggesting additional temporal structures were unmodelled. A walk-forward expanding-window procedure facilitated forecast evaluation, yielding one-, two-, and three-month-ahead forecasts along with metrics such as MAE, MSE, RMSE, and R^2 . The model was updated using 420 observations up to December 2024 and forecasted values for January-March 2025, presenting results on a Power BI dashboard.

CHAPTER 6

RESULTS

6.1 ARIMA Test-Period Performance

Table 6:1 restates the walk-forward test-period performance of the final ARIMA(2,1,2) model (Section 5.7), forming the basis for the discussion that follows. Across the historical test partition (March 2017–December 2023), the model achieved strong predictive performance at all three forecasting horizons. At the one-month horizon, the model recorded an RMSE of RM2,800.45 million and an R^2 of 0.8738, indicating that the model explains a large proportion of the variation in the test-period export values. As expected, performance declines gradually as the forecast horizon lengthens: RMSE increases from RM2,800.45 million (1-month) to RM2,971.24 million (2-month) and RM3,259.24 million (3-month), while R^2 decreases from 0.8738 to 0.8575 and 0.8266 respectively. This pattern is consistent with the nature of multi-step forecasting, in which uncertainty compounds further into the future, but the relatively modest deterioration across horizons suggests that the export series contains substantial short-term persistence that the ARIMA structure captures effectively.

6.2 Out-of-Sample Validation Discussion

The out-of-sample validation against January–March 2025 (Section 5.8) provides a practical “real-world” check on the walk-forward test performance above, since these three months were not used in either the historical test evaluation or the model identification stage. The January and March 2025 forecasts were close to the observed values, with absolute percentage errors of 2.59% and 1.73% respectively, and both observations fell within the model’s 95% forecast interval. February 2025, however, exhibited a substantially larger error of 17.47%, with the model overestimating exports and the observed value falling below the lower bound of the 95% forecast interval. Although this validation window is small ($n = 3$) and should not

be over-interpreted statistically, the pattern is consistent with the diagnostic results reported in Section 5.5: the Ljung–Box test strongly rejected the null of white-noise residuals, indicating remaining unmodelled temporal structure. The February miss is therefore interpreted as evidence that, while ARIMA(2,1,2) provides robust short-term performance on average, it can still struggle in months where the series deviates sharply from recent momentum—potentially reflecting seasonal trough effects or other short-term structural shocks that are not explicitly represented in a non-seasonal ARIMA specification.

6.3 Dashboard Results

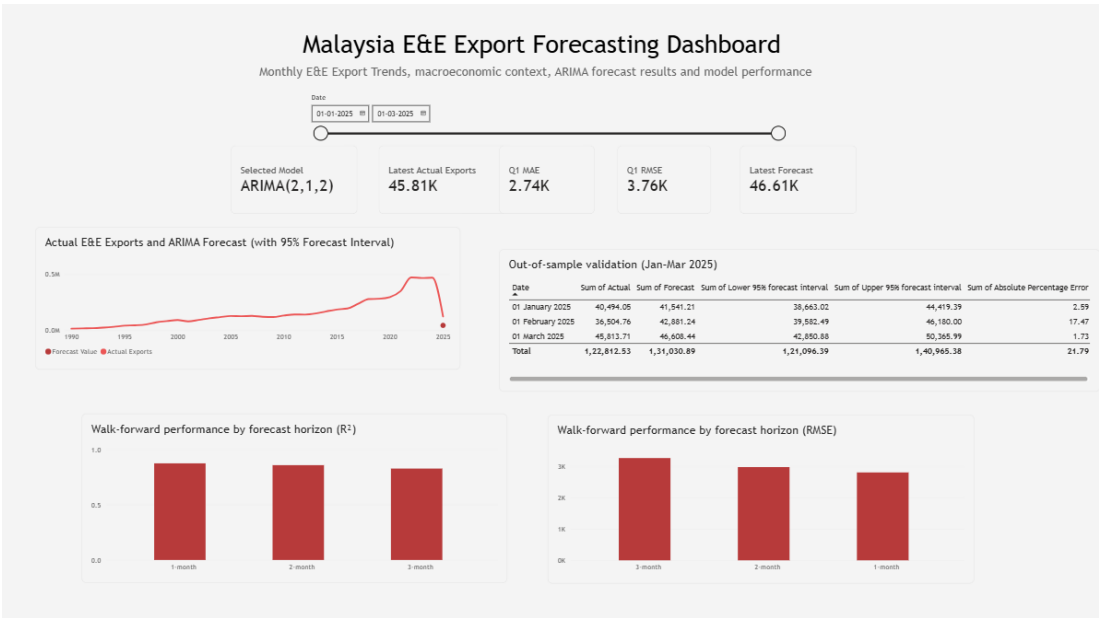


Figure 6:1 Dashboard Result

Figure 6:1 presents the completed one-page Power BI dashboard described in Section 5.9. The dashboard operationalises the third research objective by consolidating the historical export series, the ARIMA(2,1,2) forecast outputs, and the evaluation results discussed in Sections 6.1 and 6.2 into a single interface. Rather than introducing new analytical findings, it reorganises the results already reported in this chapter into the form in which they are intended to be consumed by the stakeholder groups identified in Section 1.7.

The KPI cards at the top of the dashboard establish the immediate context for interpretation. They report the selected model specification, ARIMA(2,1,2), the most recent observed E&E export value, the corresponding forecast value, and the January–March 2025 validation error metrics (MAE of RM2,739.45 million and RMSE of RM3,758.89 million). Presenting the model specification alongside its error metrics is deliberate: it ensures that any forecast read from the dashboard is accompanied by an explicit indication of the accuracy that can reasonably be expected of it, rather than being presented as a precise figure in isolation.

The central time-series chart displays the full historical series from January 1990 to March 2025 together with the January–March 2025 forecast and its 95% forecast interval. This view makes two things visible simultaneously. First, the long-run upward trajectory of Malaysia's E&E exports, together with the shock periods identified in Chapter 4, is retained as context for the forecast rather than being cropped out. Second, the shaded forecast band communicates the uncertainty attached to the projection visually, so that the width of the interval — and therefore the degree of confidence warranted — is apparent without requiring the user to interpret numerical error metrics.

The out-of-sample validation table reports the January–March 2025 actual values, point forecasts, forecast intervals, and absolute percentage errors on a month-by-month basis. Its diagnostic value is evident in the February 2025 row, where the 17.47% error and the observation falling below the lower interval bound are both directly visible. This is a deliberate design decision: the dashboard exposes the month in which the model performed poorly rather than reporting only aggregate accuracy, allowing users to see the conditions under which the forecast is less reliable. The walk-forward evaluation visuals beneath the table extend this by displaying RMSE and R^2 across the one-, two-, and three-month horizons, making the gradual degradation of accuracy with horizon length — from R^2 of 0.8738 to 0.8266 — immediately legible and supporting an informed choice of forecast horizon.

Finally, the explanatory notes panel states that the macroeconomic indicators shown are provided for descriptive economic context only and are not inputs to the

ARIMA model. This clarification is necessary because the simultaneous display of exports alongside CPI inflation, the MYR/USD exchange rate, and the U.S. Industrial Production Index could otherwise invite the inference that these variables drive the forecast. Taken as a whole, the dashboard therefore achieves what the separate tables and figures of Chapters 5 and 6 cannot: it allows the export trend, the short-term outlook, the associated uncertainty, and the model's demonstrated accuracy to be assessed together in a single view, with the limitations of the forecast made visible alongside the forecast itself.

6.4 Chapter Summary

This chapter evaluates the ARIMA (2,1,2) model's forecasting performance for Malaysia's monthly E&E exports, revealing strong results during walk-forward evaluation from March 2017 to December 2023. RMSE increased progressively from RM2,800.45 million at one month to RM3,259.24 million at three months, while R^2 gradually declined. Despite a smaller validation dataset for January–March 2025, the model demonstrated reasonable accuracy with MAE of RM2,739.45 million and RMSE of RM3,758.89 million, although February's error was notably high. The presented Power BI dashboard consolidates the results, including KPIs and forecast accuracy, making insights accessible to stakeholders. Overall, ARIMA(2,1,2) proves effective for short-term forecasting, with identified seasonal patterns highlighting areas for model improvement, to be discussed further in Chapter 7.

CHAPTER 7

CONCLUSION

7.1 Achievement of the Research Objectives

This study set out to develop and evaluate an ARIMA-based time-series forecasting model for Malaysia's monthly Electrical and Electronics (E&E) exports, using 423 monthly observations spanning January 1990 to March 2025, and to communicate the resulting forecasts through an interactive dashboard. Three objectives were defined in Section 1.4, and each has been addressed.

The first objective, to analyse the patterns, trends and relationships between Malaysia's monthly E&E export values, their time-series characteristics and the selected macroeconomic indicators, was addressed in Chapter 4. The exploratory analysis established that the series follows a sustained long-run upward trend interrupted by identifiable shock periods, and characterised its autocorrelation structure and distributional properties. The relationships between exports and CPI inflation, the MYR/USD exchange rate, and the U.S. Industrial Production Index were examined descriptively to provide economic context, without these indicators being used as model inputs.

The second objective, to develop, validate and evaluate an ARIMA model for forecasting at the one-, two- and three-month-ahead horizons, was addressed in Chapters 5 and 6. Augmented Dickey-Fuller testing confirmed that the raw training series was non-stationary ($ADF = 0.3140$, $p = 0.9780$) and became stationary after first differencing ($ADF = -6.7915$, $p < 0.0001$), establishing $d = 1$. Six candidate $ARIMA(p,1,q)$ specifications were compared using AIC, and $ARIMA(2,1,2)$ was selected on the lowest AIC value (5289.81). Walk-forward expanding-window evaluation across the March 2017–December 2023 test partition yielded an RMSE of RM2,800.45 million and R^2 of 0.8738 at one month, degrading only moderately to

RM3,259.24 million and 0.8266 at three months. Out-of-sample validation against January–March 2025 produced an MAE of RM2,739.45 million and an RMSE of RM3,758.89 million.

The third objective, to develop an interactive dashboard presenting historical trends, model performance and forecast results, was addressed in Sections 5.9 and 6.3. The one-page Power BI dashboard consolidates the KPI summary, the historical series with the forecast and its 95% interval, the out-of-sample validation table, and the walk-forward performance visuals, making the forecasting outputs directly usable by the stakeholder groups identified in Section 1.7.

The study therefore concludes that a non-seasonal ARIMA(2,1,2) specification provides a statistically sound, interpretable and operationally useful basis for short-term forecasting of Malaysia's E&E exports, with forecast reliability highest at the one-month horizon and declining only gradually out to three months.

7.2 Contributions of the Study

The study makes four contributions.

It provides the first properly identified and diagnostically validated ARIMA forecasting model for Malaysia's monthly E&E export series. Previous Malaysia-specific work either estimated long-run cointegrating relationships rather than producing operational forecasts (Wong, 2008; Muhammad and Yaacob, 2009; Hakim, 2024), or applied a SARIMA model at quarterly resolution (Sheng and Ismail, 2025). This study addresses the gap identified in Section 2.6 directly.

It demonstrates the value of monthly resolution. Retaining monthly observations yields 423 data points rather than the small number available quarterly, providing a substantially stronger basis for order identification and diagnostic testing, and producing forecasts at the horizon at which short-term trade and production decisions are actually taken.

It establishes a documented performance benchmark. By reporting evaluation metrics across three horizons under a walk-forward procedure, and validating against a genuinely unseen period, the study provides reference figures against which alternative specifications can be compared. The supplementary Random Forest, SVR and XGBoost comparison in Appendix A extends this benchmark to non-linear approaches.

It delivers the forecasts in a usable form. The Power BI dashboard translates the statistical output into an accessible interface, and its explicit treatment of forecast uncertainty and of the descriptive-only status of the macroeconomic indicators guards against the over-confident interpretation to which point forecasts are otherwise prone.

7.3 Limitations of the Study

Five limitations qualify these findings and should be taken into account when interpreting them.

Residual autocorrelation remains. The Ljung–Box test rejected the null hypothesis of white-noise residuals, indicating that temporal structure in the series is not fully captured by the selected specification. The forecasts are therefore statistically usable but not optimal, and the reported intervals may understate true forecast uncertainty.

Seasonality was not explicitly modelled. The final model is non-seasonal. The February 2025 overprediction of 17.47%, falling outside the 95% interval, is consistent with an unmodelled seasonal trough and represents the clearest observed consequence of this choice.

The out-of-sample validation window is small. Only three months (January–March 2025) were available as genuinely unseen data. With $n = 3$, the reported out-of-sample MAE and RMSE are indicative rather than statistically robust, and a single large error exerts disproportionate influence on them.

The model is univariate. ARIMA forecasts the series solely from its own history. It cannot anticipate turning points driven by external events — trade policy changes, global demand shocks, or semiconductor cycle reversals — and will lag rather than lead such structural breaks.

No stakeholder evaluation was conducted. The usefulness of the dashboard to the four groups identified in Section 1.7 is inferred from their documented information needs rather than established through consultation or usability testing, so claims about its practical value remain untested.

7.4 Recommendations for Future Work

The limitations above point to five directions for further research.

Fit and evaluate a seasonal specification. A $SARIMA(p,d,q)(P,D,Q)_{12}$ model should be estimated on the same data and compared against the $ARIMA(2,1,2)$ benchmark reported here. Given both the residual diagnostics and the February 2025 deviation, this is the single most promising avenue for improving accuracy, and the benchmark figures in Chapter 6 provide a direct basis for comparison.

Extend the validation window. As additional monthly observations accumulate beyond March 2025, the out-of-sample evaluation should be repeated over a window of twelve months or more, allowing performance to be assessed across a full seasonal cycle rather than a three-month fragment.

Explore multivariate and hybrid approaches. $ARIMAX$ or $SARIMAX$ specifications incorporating leading indicators such as the U.S. Industrial Production Index or global semiconductor billings could be tested, subject to the constraint that any exogenous variable must itself be forecast or assumed at future time points. Hybrid $ARIMA$ –machine-learning designs, in which a non-linear model is fitted to the $ARIMA$ residuals, offer a further route to capturing the structure the linear model leaves behind, building on the supplementary comparison in Appendix A.

Model structural breaks explicitly. Intervention analysis or regime-switching specifications could account for the shock periods identified in Chapter 4, such as the 2008–2009 global financial crisis and the 2020 pandemic disruption, which a standard ARIMA treats as ordinary variation.

Evaluate the dashboard with its intended users. Usability testing or structured interviews with policymakers, exporters, and analysts would establish whether the forecasts and uncertainty information are interpreted as intended, and would allow the interface to be refined against actual decision-making practice. Automating the data refresh and model refit would further extend the dashboard from a static deliverable into an operational monitoring tool.

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